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REPORTING CROSS-LINK INTERFERENCE ASSOCIATED WITH AN ELECTROMAGNETIC RADIATION REFLECTION RELAY SERVICE

Abstract

Various aspects of the present disclosure generally relate to wireless communication. In some aspects, a first network node may receive a signal comprising a first signal characteristic associated with a first pre-coding vector corresponding to a first communication channel between the first network node and a second network node and a second signal characteristic associated with a second pre-coding vector corresponding to a second communication channel between the first network node and an electromagnetic radiation reflection relay network node. The first network node may transmit channel estimation information corresponding to the signal. Numerous other aspects are described.

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Background/Summary

FIELD OF THE DISCLOSURE

[0001] Aspects of the present disclosure generally relate to wireless communication and to techniques and apparatuses for reporting cross-link interference associated with an electromagnetic radiation reflection relay service.

INTRODUCTION

[0002] Wireless communication systems are widely deployed to provide various telecommunication services such as telephony, video, data, messaging, and broadcasts. Typical wireless communication systems may employ multiple-access technologies capable of supporting communication with multiple users by sharing available system resources (e.g., bandwidth, transmit power, or the like). Examples of such multiple-access technologies include code division multiple access (CDMA) systems, time division multiple access (TDMA) systems, frequency division multiple access (FDMA) systems, orthogonal frequency division multiple access (OFDMA) systems, single-carrier frequency division multiple access (SC-FDMA) systems, time division synchronous code division multiple access (TD-SCDMA) systems, and Long Term Evolution (LTE). LTE/LTE-Advanced is a set of enhancements to the Universal Mobile Telecommunications System (UMTS) mobile standard promulgated by the Third Generation Partnership Project (3GPP).

[0003] A wireless network may include one or more base stations that support communication for a user equipment (UE) or multiple UEs. A UE may communicate with a base station via downlink communications and uplink communications. “Downlink” (or “DL”) refers to a communication link from the base station to the UE, and “uplink” (or “UL”) refers to a communication link from the UE to the base station.

[0004] The above multiple access technologies have been adopted in various telecommunication standards to provide a common protocol that enables different UEs to communicate on a municipal, national, regional, and/or global level. New Radio (NR), which may be referred to as 5G, is a set of enhancements to the LTE mobile standard promulgated by the 3GPP. NR is designed to better support mobile broadband internet access by improving spectral efficiency, lowering costs, improving services, making use of new spectrum, and better integrating with other open standards using orthogonal frequency division multiplexing (OFDM) with a cyclic prefix (CP) (CP-OFDM) on the downlink, using CP-OFDM and/or single-carrier frequency division multiplexing (SC-FDM) (also known as discrete Fourier transform spread OFDM (DFT-s-OFDM)) on the uplink, as well as supporting beamforming, multiple-input multiple-output (MIMO) antenna technology, and carrier aggregation. As the demand for mobile broadband access continues to increase, further improvements in LTE, NR, and other radio access technologies remain useful.

SUMMARY

[0005] Some aspects described herein relate to a first network node for wireless communication. The first network node may include a memory and one or more processors coupled to the memory. The one or more processors may be configured to receive a signal comprising a first signal characteristic associated with a first pre-coding vector corresponding to a first communication channel between the first network node and a second network node and a second signal

characteristic associated with a second pre-coding vector corresponding to a second communication channel between the first network node and an electromagnetic radiation reflection relay network node. The one or more processors may be configured to transmit channel estimation information corresponding to the signal, wherein the channel estimation information includes first channel estimation information and second channel estimation information, wherein the first channel estimation information corresponds to the first communication channel and is based on the first pre-coding vector, and wherein the second channel estimation information corresponds to the second communication channel and is based on a second pre-coding vector associated with the second communication channel

[0006] Some aspects described herein relate to a first network node for wireless communication. The first network node may include a memory and one or more processors coupled to the memory. The one or more processors may be configured to transmit cross-link interference (CLI) configuration information that includes first information indicative of a reference signal resource. The one or more processors may be configured to receive channel estimation information corresponding to a signal that is based on the reference signal resource, wherein the channel estimation information includes first channel estimation information and second channel estimation information, wherein the first channel estimation information corresponds to a first communication channel between a second network node and a third network node and is based on a first pre-coding vector associated with the first communication channel, and wherein the second channel estimation information corresponds to a second communication channel between an electromagnetic radiation reflection relay network node and the third network node and is based on a second pre-coding vector associated with the second communication channel. The one or more processors may be configured to transmit reflection relay configuration information to the electromagnetic radiation reflection relay network node, wherein the reflection relay configuration information is indicative of one or more electromagnetic reflection parameters based on the channel estimation information.

[0007] Some aspects described herein relate to a first network node for wireless communication. The first network node may include a memory and one or more processors coupled to the memory. The one or more processors may be configured to receive, from an aggressor network node, a plurality of aggressor reference signals, wherein the plurality of aggressor reference signals includes a first aggressor reference signal pre-coded by first pre-coding information and a second aggressor reference signal pre-coded by second pre-coding information. The one or more processors may be configured to receive, from an electromagnetic radiation reflection relay network node, a plurality of reflected reference signals, wherein the plurality of reflected reference signals includes a first reflected reference signal pre-coded by third pre-coding information and a second reflected reference signal pre-coded by fourth pre-coding information, wherein the first aggressor reference signal and the first reflected reference signal overlap in time, and the second aggressor reference signal and the second reflected reference signal overlap in time. The one or more processors may be configured to transmit, to a second network node, channel estimation information including first channel estimation information and second channel estimation information, wherein the channel estimation information is based on the plurality of aggressor reference signals and the plurality of reflected reference signals, wherein the first channel information is indicative of one or more channel characteristics corresponding to a first channel between the first network node and the aggressor network node, and wherein the second channel information is indicative of one or more channel characteristics corresponding to a second channel between the first network node and the electromagnetic radiation reflection relay network node.

[0008] Some aspects described herein relate to a method of wireless communication performed by a first network node. The method may include receiving a signal comprising a first signal characteristic associated with a first pre-coding vector corresponding to a first communication channel between the first network node and a second network node and a second signal characteristic associated with a second pre-coding vector corresponding to a second communication channel between the first network node and an electromagnetic radiation reflection relay network node. The method may include transmitting channel estimation information corresponding to the signal, wherein the channel estimation information includes first channel estimation information and second channel estimation information, wherein the first channel estimation information corresponds to the first communication channel and is based on the first pre-coding vector, and wherein the second channel estimation information corresponds to the second communication channel and is based on a second pre-coding vector associated with the second communication channel.

[0009] Some aspects described herein relate to a method of wireless communication performed by a first network node. The method may include transmitting CLI configuration information that includes first information indicative of a reference signal resource. The method may include receiving channel estimation information corresponding to a signal that is based on the reference signal resource, wherein the channel estimation information includes first channel estimation information and second channel estimation information, wherein the first channel estimation information corresponds to a first communication channel between a second network node and a third network node and is based on a first pre-coding vector associated with the first communication channel, and wherein the second channel estimation information corresponds to a second communication channel between an electromagnetic radiation reflection relay network node and the third network node and is based on a second pre-coding vector associated with the second communication channel. The method may include transmitting reflection relay configuration information to the electromagnetic radiation reflection relay network node, wherein the reflection relay configuration information is indicative of one or more electromagnetic reflection parameters based on the channel estimation information.

[0010] Some aspects described herein relate to a method of wireless communication performed by a first network node. The method may include receiving, from an aggressor network node, a plurality of aggressor reference signals, wherein the plurality of aggressor reference signals includes a first aggressor reference signal pre-coded by first pre-coding information and a second aggressor reference signal pre-coded by second pre-coding information. The method may include receiving, from an electromagnetic radiation reflection relay network node, a plurality of reflected reference signals, wherein the plurality of reflected reference signals includes a first reflected reference signal pre-coded by third pre-coding information and a second reflected reference signal pre-coded by fourth pre-coding information, wherein the first aggressor reference signal and the first reflected reference signal overlap in time, and the second aggressor reference signal and the second reflected reference signal overlap in time. The method may include transmitting, to a second network node, channel estimation information including first channel estimation information and second channel estimation information, wherein the channel estimation information is based on the plurality of aggressor reference signals and the plurality of reflected reference signals, wherein the first channel information is indicative of one or more channel characteristics corresponding to a first channel between the first network node and the aggressor network node, and wherein the second channel information is indicative of one or more channel characteristics corresponding to a second channel between the first network node and the electromagnetic radiation reflection relay network node.

[0011] Some aspects described herein relate to a non-transitory computer-readable medium that stores a set of instructions for wireless communication by a first network node. The set of instructions, when executed by one or more processors of the first network node, may cause the first network node to receive a signal comprising a first signal characteristic associated with a first pre-coding vector corresponding to a first communication channel between the first network node and a second network node and a second signal characteristic associated with a second pre-coding vector corresponding to a second communication channel between the first network node and an electromagnetic radiation reflection relay network node. The set of instructions, when executed by one or more processors of the first network node, may cause the first network node to transmit channel estimation information corresponding to the signal, wherein the channel estimation information includes first channel estimation information and second channel estimation information, wherein the first channel estimation information corresponds to the first communication channel and is based on the first pre-coding vector, and wherein the second channel estimation information corresponds to the second communication channel and is based on a second pre-coding vector associated with the second communication channel.

[0012] Some aspects described herein relate to a non-transitory computer-readable medium that stores a set of instructions for wireless communication by a first network node. The set of instructions, when executed by one or more processors of the first network node, may cause the first network node to transmit CLI configuration information that includes first information indicative of a reference signal resource. The set of instructions, when executed by one or more processors of the first network node, may cause the first network node to receive channel estimation information corresponding to a signal that is based on the reference signal resource, wherein the channel estimation information includes first channel estimation information and second channel estimation information, wherein the first channel estimation information corresponds to a first communication channel between a second network node and a third network node and is based on a first pre-coding vector associated with the first communication channel, and wherein the second channel estimation information corresponds to a second communication channel between an electromagnetic radiation reflection relay network node and the third network node and is based on a second pre-coding vector associated with the second communication channel. The set of instructions, when executed by one or more processors of the first network node, may cause the first network node to transmit reflection relay configuration information to the electromagnetic radiation reflection relay network node, wherein the reflection relay configuration information is indicative of one or more electromagnetic reflection parameters based on the channel estimation information.

[0013] Some aspects described herein relate to a non-transitory computer-readable medium that stores a set of instructions for wireless communication by a first network node. The set of instructions, when executed by one or more processors of the first network node, may cause the first network node to receive, from an aggressor network node, a plurality of aggressor reference signals, wherein the plurality of aggressor reference signals includes a first aggressor reference signal pre-coded by first pre-coding information and a second aggressor reference signal pre-coded by second pre-coding information. The set of instructions, when executed by one or more processors of the first network node, may cause the first network node to receive, from an electromagnetic radiation reflection relay network node, a plurality of reflected reference signals, wherein the plurality of reflected reference signals includes a first reflected reference signal pre-coded by third pre-coding information and a second reflected reference signal pre-coded by fourth pre-coding information, wherein the first aggressor reference signal and the first reflected reference signal overlap in time, and the second aggressor reference signal and the second reflected reference signal overlap in time. The set of instructions, when executed by one or more processors of the first network node, may cause the first network node to transmit, to a second network node, channel estimation information including first channel estimation information and second channel estimation information, wherein the channel estimation information is based on the plurality of aggressor reference signals and the plurality of reflected reference signals, wherein the first channel information is indicative of one or more channel characteristics corresponding to a first channel between the first network node and the aggressor network node, and wherein the second channel information is indicative of one or more channel characteristics corresponding to a second channel between the first network node and the electromagnetic radiation reflection relay network node.

[0014] Some aspects described herein relate to an apparatus for wireless communication. The apparatus may include means for receiving a signal comprising a first signal characteristic associated with a first pre-coding vector corresponding to a first communication channel between the apparatus and a first network node and a second signal characteristic associated with a second pre-coding vector corresponding to a second communication channel between the apparatus and an electromagnetic radiation reflection relay network node. The apparatus may include means for transmitting channel estimation information corresponding to the signal, wherein the channel estimation information includes first channel estimation information and second channel estimation information, wherein the first channel estimation information corresponds to the first communication channel and is based on the first pre-coding vector, and wherein the second channel estimation information corresponds to the second communication channel and is based on a second pre-coding vector associated with the second communication channel.

[0015] Some aspects described herein relate to an apparatus for wireless communication. The apparatus may include means for transmitting CLI configuration information that includes first information indicative of a reference signal resource. The apparatus may include means for receiving channel estimation information corresponding to a signal that is based on the reference signal resource, wherein the channel estimation information includes first channel estimation information and second channel estimation information, wherein the first channel estimation information corresponds to a first communication channel between a first network node and a second network node and is based on a first pre-coding vector associated with the first communication channel, and wherein the second channel estimation information corresponds to a second communication channel between an electromagnetic radiation reflection relay network node and the second network node and is based on a second pre-coding vector associated with the second communication channel. The apparatus may include means for transmitting reflection relay configuration information to the electromagnetic radiation reflection relay network node, wherein the reflection relay configuration information is indicative of one or more electromagnetic reflection parameters based on the channel estimation information.

[0016] Some aspects described herein relate to an apparatus for wireless communication. The apparatus may include means for receiving, from an aggressor network node, a plurality of aggressor reference signals, wherein the plurality of aggressor reference signals includes a first aggressor reference signal pre-coded by first pre-coding information and a second aggressor reference signal pre-coded by second pre-coding information. The apparatus may include means for receiving, from an electromagnetic radiation reflection relay network node, a plurality of reflected reference signals, wherein the plurality of reflected reference signals includes a first reflected reference signal pre-coded by third pre-coding information and a second reflected reference signal pre-coded by fourth pre-coding information, wherein the first aggressor reference signal and the first reflected reference signal overlap in time, and the second aggressor reference signal and the second reflected reference signal overlap in time. The apparatus may include means for transmitting, to a first network node, channel estimation information including first channel estimation information and second channel estimation information, wherein the channel estimation information is based on the plurality of aggressor reference signals and the plurality of reflected reference signals, wherein the first channel information is indicative of one or more channel characteristics corresponding to a first channel between the apparatus and the aggressor network node, and wherein the second channel information is indicative of one or more channel characteristics corresponding to a second channel between the apparatus and the electromagnetic radiation reflection relay network node.

[0017] Aspects generally include a method, apparatus, system, computer program product, non-transitory computer-readable medium, user equipment, base station, wireless communication device, and/or processing system as substantially described herein with reference to and as illustrated by the drawings and specification.

[0018] The foregoing has outlined rather broadly the features and technical advantages of examples according to the disclosure in order that the detailed description that follows may be better

understood. Additional features and advantages will be described hereinafter. The conception and specific examples disclosed may be readily utilized as a basis for modifying or designing other structures for carrying out the same purposes of the present disclosure. Such equivalent constructions do not depart from the scope of the appended claims. Characteristics of the concepts disclosed herein, both their organization and method of operation, together with associated advantages, will be better understood from the following description when considered in connection with the accompanying figures. Each of the figures is provided for the purposes of illustration and description, and not as a definition of the limits of the claims.
[0019] While aspects are described in the present disclosure by illustration to some examples, those skilled in the art will understand that such aspects may be implemented in many different arrangements and scenarios. Techniques described herein may be implemented using different platform types, devices, systems, shapes, sizes, and/or packaging arrangements. For example, some aspects may be implemented via integrated chip embodiments or other non-module-component based devices (e.g., end-user devices, vehicles, communication devices, computing devices, industrial equipment, retail/purchasing devices, medical devices, and/or artificial intelligence devices). Aspects may be implemented in chip-level components, modular components, non-modular components, non-chip-level components, device-level components, and/or system-level components. Devices incorporating described aspects and features may include additional components and features for implementation and practice of claimed and described aspects. For example, transmission and reception of wireless signals may include one or more components for analog and digital purposes (e.g., hardware components including antennas, radio frequency (RF) chains, power amplifiers, modulators, buffers, processors, interleavers, adders, and/or summers). It is intended that aspects described herein may be practiced in a wide variety of devices, components, systems, distributed arrangements, and/or end-user devices of varying size, shape, and constitution.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0020] So that the above-recited features of the present disclosure can be understood in detail, a more particular description, briefly summarized above, may be had by reference to aspects, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only certain typical aspects of this disclosure and are therefore not to be considered limiting of its scope, for the description may admit to other equally effective aspects. The same reference numbers in different drawings may identify the same or similar elements.
[0021] FIG. **1** is a diagram illustrating an example environment, in accordance with the present disclosure.
[0022] FIG. **2** is a diagram illustrating example components of an apparatus, in accordance with the present disclosure.
[0023] FIG. **3** is a diagram illustrating an example of a wireless network, in accordance with the present disclosure.
[0024] FIG. **4** is a diagram illustrating an environment including a network node in wireless communication with another network node, in accordance with the present disclosure.
[0025] FIG. **5** is a diagram illustrating an example of an open-radio access network architecture, in accordance with the present disclosure.
[0026] FIG. **6** is a diagram illustrating an example of wireless communication, in accordance with the present disclosure.
[0027] FIG. **7** is a diagram illustrating an example associated with reporting cross-link interference (CLI) associated with an electromagnetic radiation reflection relay service, in accordance with the

present disclosure.
[0028] FIGS. **8-10** are diagrams illustrating example processes associated with reporting CLI associated with an electromagnetic radiation reflection relay service, in accordance with the present disclosure.
[0029] FIG. **11** is a diagram of an example apparatus for wireless communication, in accordance with the present disclosure.
DETAILED DESCRIPTION
[0030] Various aspects of the disclosure are described more fully hereinafter with reference to the accompanying drawings. This disclosure may, however, be embodied in many different forms and should not be construed as limited to any specific structure or function presented throughout this disclosure. Rather, these aspects are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the disclosure to those skilled in the art. One skilled in the art should appreciate that the scope of the disclosure is intended to cover any aspect of the disclosure disclosed herein, whether implemented independently of or combined with any other aspect of the disclosure. For example, an apparatus may be implemented or a method may be practiced using any number of the aspects set forth herein. In addition, the scope of the disclosure is intended to cover such an apparatus or method which is practiced using other structure, functionality, or structure and functionality in addition to or other than the various aspects of the disclosure set forth herein. It should be understood that any aspect of the disclosure disclosed herein may be embodied by one or more elements of a claim.
[0031] Aspects and examples generally include a method, apparatus, network node, system, computer program product, non-transitory computer-readable medium, user equipment, base station, wireless communication device, and/or processing system as described or substantially described herein with reference to and as illustrated by the drawings and specification.
[0032] This disclosure may be readily utilized as a basis for modifying or designing other structures for carrying out the same purposes of the present disclosure. Such equivalent constructions do not depart from the scope of the appended claims. Characteristics of the concepts disclosed herein, both their organization and method of operation, together with associated advantages, are better understood from the following description when considered in connection with the accompanying figures. Each of the figures is provided for the purposes of illustration and description, and not as a definition of the limits of the claims.
[0033] While aspects are described in the present disclosure by illustration to some examples, such aspects may be implemented in many different arrangements and scenarios. Techniques described herein may be implemented using different platform types, devices, systems, shapes, sizes, and/or packaging arrangements. For example, some aspects may be implemented via integrated chip embodiments or other non-module-component-based devices (e.g., end-user devices, vehicles, communication devices, computing devices, industrial equipment, retail/purchasing devices, medical devices, and/or artificial intelligence devices). Aspects may be implemented in chip-level components, modular components, non-modular components, non-chip-level components, device-level components, and/or system-level components. Devices incorporating described aspects and features may include additional components and features for implementation and practice of claimed and described aspects. For example, transmission and reception of wireless signals may include one or more components for analog and digital purposes (e.g., hardware components including antennas, radio frequency (RF) chains, power amplifiers, modulators, buffers, processors, interleavers, adders, and/or summers). Aspects described herein may be practiced in a wide variety of devices, components, systems, distributed arrangements, and/or end-user devices of varying size, shape, and constitution.
[0034] Several aspects of telecommunication systems will now be presented with reference to various apparatuses and techniques. These apparatuses and techniques will be described in the following detailed description and illustrated in the accompanying drawings by various blocks,

modules, components, circuits, steps, processes, algorithms, or the like (collectively referred to as "elements"). These elements may be implemented using hardware, software, or combinations thereof. Whether such elements are implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system.

[0035] FIG. **1** is a diagram illustrating an example environment **100** in which apparatuses and/or methods described herein may be implemented, in accordance with the present disclosure. As shown in FIG. **1**, the environment **100** may include a network node **102**, a network node **104**, and a network node **106** that may communicate with one another via a network **108**. The network nodes **102**, **104**, and **106** may be dispersed throughout the network **108**, and each network node **102**, **104**, and **106** may be stationary and/or mobile. The network **108** may include wired communication connections, wireless communication connections, or a combination of wired and wireless communication connections.

[0036] The network **108** may include, for example, a cellular network (e.g., a Long-Term Evolution (LTE) network, a code division multiple access (CDMA) network, a 3G network, a 4G network, a 5G network, another type of next generation network, and/or the like), a public land mobile network (PLMN), a local area network (LAN), a wide area network (WAN), a metropolitan area network (MAN), a telephone network (e.g., the Public Switched Telephone Network (PSTN)), a private network, an ad hoc network, an intranet, the Internet, a fiber optic-based network, a cloud computing network, or the like, and/or a combination of these or other types of networks.

[0037] In general, any number of networks **108** may be deployed in a given geographic area. Each network **108** may support a particular radio access technology (RAT) and may operate on one or more frequencies. A RAT may be referred to as a radio technology, an air interface, or the like. A frequency may be referred to as a carrier, a frequency channel, or the like. Each frequency may support a single RAT in a given geographic area in order to avoid interference between wireless networks of different RATs. In some cases, Open-RAT, New Radio (NR) or 5G RAT networks may be deployed.

[0038] In some aspects, the environment **100** may include one or more non-terrestrial network (NTN) deployments in which a non-terrestrial wireless communication device may include a network node. The network node may include a UE (which may be referred to herein, interchangeably, as a "non-terrestrial UE"), a base station (referred to herein, interchangeably, as a "non-terrestrial BS" and "non-terrestrial base station"), and/or a relay station (referred to herein, interchangeably, as a "non-terrestrial relay station"), among other examples. As used herein, "NTN" may refer to a network for which access is facilitated by a non-terrestrial UE, non-terrestrial base station, and/or a non-terrestrial relay station, among other examples.

[0039] One or more of the network nodes **102**, **104**, and **106** may be, include, of be included in any number of non-terrestrial wireless communication devices. A non-terrestrial wireless communication device may include a satellite, a manned aircraft system, an unmanned aircraft system (UAS) platform, and/or the like. A satellite may include a low-earth orbit (LEO) satellite, a medium-earth orbit (MEO) satellite, a geostationary earth orbit (GEO) satellite, a high elliptical orbit (HEO) satellite, and/or the like. A manned aircraft system may include an airplane, helicopter, a dirigible, and/or the like. A UAS platform may include a high-altitude platform station (HAPS), and may include a balloon, a dirigible, an airplane, and/or the like. Satellites may communicate directly and/or indirectly with other entities in the environment using satellite communication. The other entities may include UEs (e.g., terrestrial UEs and/or non-terrestrial UEs), other satellites in the one or more NTN deployments, other types of base stations (e.g., stationary and/or ground-based BSs), relay stations, and/or one or more components and/or devices included in a core network, among other examples.

[0040] As described herein, a node (which may be referred to as a node, a network node, a network entity, or a wireless node) may include, be, or be included in (e.g., be a component of) a base station (e.g., any base station described herein), a UE (e.g., any UE described herein), a network controller, an apparatus, a device, a computing system, an integrated access and backhauling (IAB) node, a distributed unit (DU), a central unit (CU), a remote/radio unit (RU) (which may also be referred to as a remote radio unit (RRU)), and/or another processing entity configured to perform any of the techniques described herein. For example, a network node may be a UE. As another example, a network node may be a base station or network entity. As another example, a first network node may be configured to communicate with a second network node or a third network node. In one aspect of this example, the first network node may be a UE, the second network node may be a base station, and the third network node may be a UE. In another aspect of this example, the first network node may be a UE, the second network node may be a base station, and the third network node may be a base station. In yet other aspects of this example, the first, second, and third network nodes may be different relative to these examples. Similarly, reference to a UE, base station, apparatus, device, computing system, or the like may include disclosure of the UE, base station, apparatus, device, computing system, or the like being a network node. For example, disclosure that a UE is configured to receive information from a base station also discloses that a first network node is configured to receive information from a second network node. Consistent with this disclosure, once a specific example is broadened in accordance with this disclosure (e.g., a UE is configured to receive information from a base station also discloses that a first network node is configured to receive information from a second network node), the broader example of the narrower example may be interpreted in the reverse, but in a broad open-ended way. In the example above where a UE is configured to receive information from a base station also discloses that a first network node is configured to receive information from a second network node, the first network node may refer to a first UE, a first base station, a first apparatus, a first device, a first computing system, a first set of one or more one or more components, a first processing entity, or the like configured to receive the information; and the second network node may refer to a second UE, a second base station, a second apparatus, a second device, a second computing system, a second set of one or more components, a second processing entity, or the like.

[0041] As described herein, communication of information (e.g., any information, signal, or the like) may be described in various aspects using different terminology. Disclosure of one communication term includes disclosure of other communication terms. For example, a first network node may be described as being configured to transmit information to a second network node. In this example and consistent with this disclosure, disclosure that the first network node is configured to transmit information to the second network node includes disclosure that the first network node is configured to provide, send, output, communicate, or transmit information to the second network node. Similarly, in this example and consistent with this disclosure, disclosure that the first network node is configured to transmit information to the second network node includes disclosure that the second network node is configured to receive, obtain, or decode the information that is provided, sent, output, communicated, or transmitted by the first network node

[0042] As shown, the network node **102** may include a communication manager **110** and a transceiver **112**. The communication manager **110** may be configured to perform one or more communication tasks as described herein. In some aspects, the communication manager **110** may direct the transceiver **112** to perform one or more communication tasks as described herein. Similarly, the network node **106** may include a communication manager **114** and a transceiver **116**. The communication manager **114** may be configured to perform one or more communication tasks as described herein. In some aspects, the communication manager **114** may direct the transceiver **116** to perform one or more communication tasks as described herein.

[0043] In some aspects, a first network node may include a communication manager **110** or a communication manager **114**. As described in more detail elsewhere herein, the communication manager **110** or **114** may receive a signal comprising a first signal characteristic associated with a first pre-coding vector corresponding to a first communication channel between the first network node and a second network node and a second signal characteristic associated with a second pre-coding vector corresponding to a second communication channel between the first network node and an electromagnetic radiation reflection relay network node; and transmit channel estimation information corresponding to the signal, wherein the channel estimation information includes first channel estimation information and second channel estimation information, wherein the first channel estimation information corresponds to the first communication channel and is based on the first pre-coding vector, and wherein the second channel estimation information corresponds to the second communication channel and is based on a second pre-coding vector associated with the second communication channel.

[0044] As described in more detail elsewhere herein, the communication manager **110** or **114** may transmit cross-link interference (CLI) configuration information that includes first information indicative of a reference signal resource; receive channel estimation information corresponding to a signal that is based on the reference signal resource, wherein the channel estimation information includes first channel estimation information and second channel estimation information, wherein the first channel estimation information corresponds to a first communication channel between a second network node and a third network node and is based on a first pre-coding vector associated with the first communication channel, and wherein the second channel estimation information corresponds to a second communication channel between an electromagnetic radiation reflection relay network node and the third network node and is based on a second pre-coding vector associated with the second communication channel; and transmit reflection relay configuration information to the electromagnetic radiation reflection relay network node, wherein the reflection relay configuration information is indicative of one or more electromagnetic reflection parameters based on the channel estimation information.

[0045] As described in more detail elsewhere herein, the communication manager **110** or **114** may receive, from an aggressor network node, a plurality of aggressor reference signals, wherein the plurality of aggressor reference signals includes a first aggressor reference signal pre-coded by first pre-coding information and a second aggressor reference signal pre-coded by second pre-coding information; receive, from an electromagnetic radiation reflection relay network node, a plurality of reflected reference signals, wherein the plurality of reflected reference signals includes a first reflected reference signal pre-coded by third pre-coding information and a second reflected reference signal pre-coded by fourth pre-coding information, wherein the first aggressor reference signal and the first reflected reference signal overlap in time, and the second aggressor reference signal and the second reflected reference signal overlap in time; and transmit, to a second network node, channel estimation information including first channel estimation information and second channel estimation information, wherein the channel estimation information is based on the plurality of aggressor reference signals and the plurality of reflected reference signals, wherein the first channel information is indicative of one or more channel characteristics corresponding to a first channel between the first network node and the aggressor network node, and wherein the second channel information is indicative of one or more channel characteristics corresponding to a second channel between the first network node and the electromagnetic radiation reflection relay network node. Additionally, or alternatively, the communication manager **110** and/or **114** may perform one or more other operations described herein.

[0046] The number and arrangement of entities shown in FIG. **1** are provided as one or more examples. In practice, there may be additional network nodes and/or networks, fewer network nodes and/or networks, different network nodes and/or networks, or differently arranged network nodes and/or networks than those shown in FIG. **1**. Furthermore, the network node **102**, **104**, and/or **106** may be implemented using a single apparatus or multiple apparatuses.

[0047] FIG. **2** is a diagram of example components of an apparatus **200**. The apparatus **200** may correspond to the network node **102** and/or the network node **106**. Additionally, or alternatively, the network node **102** and/or the network node **106** may include one or more apparatuses **200** and/or one or more components of the apparatus **200**. For example, in some aspects, the apparatus **200** may include an apparatus (e.g., a device, a device component, a modem, a chip, and/or a set of device components, among other examples) that is configured to perform a wireless communication method at a network node, as described herein. As shown in FIG. **2**, the apparatus **200** may include components such as a bus **205**, a processor **210**, a memory **215**, an input component **220**, an output component **225**, a communication interface **230**, a communication manager **235**, and a reference signal processing component **240**. Any one or more of the components **205**, **210**, **215**, **220**, **225** **230**, **235**, and/or **240** may be implemented in hardware, software, or a combination of hardware and software.

[0048] The bus **205** includes a component that permits communication among the components of the apparatus **200**. The processor **210** includes a central processing unit (CPU), a graphics processing unit (GPU), an accelerated processing unit (APU), a digital signal processor (DSP), a microprocessor, a microcontroller, a field-programmable gate array (FPGA), an application-specific integrated circuit (ASIC), and/or another type of processing component. In some aspects, the processor **210** includes one or more processors capable of being programmed to perform a function.

[0049] The memory **215** includes a random-access memory (RAM), a read only memory (ROM), and/or another type of dynamic or static storage device (e.g., a flash memory, a magnetic memory, and/or an optical memory) that stores information and/or instructions for use by the processor **210**. The memory **215** may store other information and/or software related to the operation and use of the apparatus **200**. For example, the memory **215** may include a hard disk (e.g., a magnetic disk, an optical disk, a magneto-optic disk, and/or a solid-state disk), a compact disc (CD), a digital versatile disc (DVD), a floppy disk, a cartridge, a magnetic tape, and/or another type of non-transitory computer-readable medium.

[0050] The input component **220** includes a component that permits the apparatus **200** to receive information, such as via user input. For example, the input component **220** may be associated with a user interface as described herein (e.g., to permit a user to interact with the one or more features of the apparatus **200**). The input component **220** may include a capacitive touchscreen display that can receive user inputs. The input component **220** may include a keyboard, a keypad, a mouse, a button, a switch, and/or a microphone, among other examples. Additionally, or alternatively, the input component **220** may include a sensor for sensing information (e.g., a vision sensor, a location sensor, an accelerometer, a gyroscope, and/or an actuator, among other examples). In some aspects, the input component **220** may include a camera (e.g., a high-resolution camera and/or a low-resolution camera, among other examples). The output component **225** may include a component that provides output from the apparatus **200** (e.g., a display, a speaker, and/or one or more light-emitting diodes (LEDs), among other examples).

[0051] The communication interface **230** may include a transmission component and/or a reception component. For example, the communication interface **230** may include a transceiver and/or one or more separate receivers and/or transmitters that enable the apparatus **200** to communicate with other devices, such as via a wired connection, a wireless connection, or a combination of wired and wireless connections. In some aspects, the communication interface may include one or more radio frequency reflective elements and/or one or more radio frequency refractive elements. The communication interface **230** may permit the apparatus **200** to receive information from another apparatus and/or provide information to another apparatus. For example, the communication interface **230** may include an Ethernet interface, an optical interface, a coaxial interface, an infrared interface, an RF interface, a universal serial bus (USB) interface, a Wi-Fi interface, a cellular network interface, a wireless modem, an inter-integrated circuit (I.sup.2C), and/or a serial peripheral interface (SPI), among other examples.

[0052] The communication manager **235** may include hardware, software, or a combination of hardware and software configured to cause the apparatus **200** to perform one or more communication tasks associated with the communication manager **110**, the transceiver **112**, the communication manager **114**, and/or the transceiver **116**. In some aspects, the communication manager **235** may be, be similar to, include, or be included in, the communication manager **110** and/or the communication manager **114** depicted in FIG. **1**. In some aspects, the communication manager **235** may include the processor **210**, the memory **215**, the input component **220**, the output component **225**, the communication interface **230**, and/or the reference signal processing component **240**, and/or one or more aspects thereof.

[0053] As indicated above, FIG. **2** is provided as an example. Other examples may differ from what is described with regard to FIG. **2**.

[0054] As described above, in some aspects, the network **108** depicted in FIG. **1** may include a cellular network that includes a RAT. While some aspects may be described herein using terminology commonly associated with a 5G or NR RAT, aspects of the present disclosure can be applied to other RATs, such as a 3G RAT, a 4G RAT, and/or a RAT subsequent to 5G (e.g., 6G).

[0055] FIG. **3** is a diagram illustrating an example of a wireless network **300**, in accordance with the present disclosure. The wireless network **300** may be or may include elements of a 4G (e.g., LTE) network, a 5G (e.g., NR) network, and/or a 6G network, among other examples. The wireless network **300** may include one or more base stations **310** (illustrated individually as a BS **310***a*, a BS **310***b*, a BS **310***c*, and a BS **310***d*), a UE **320** or multiple UEs **320** (illustrated individually as a UE **320***a*, a UE **320***b*, a UE **320***c*, a UE **320***d*, and a UE **320***e*), and/or other network entities. A base station **310** is an entity that communicates with UEs **320**. A base station **310** may include, for example, an NR base station, an LTE base station, a Node B, an eNB (e.g., in 4G), a gNB (e.g., in 5G), an access point, and/or a transmission reception point (TRP). Each base station **310** may provide communication coverage for a particular geographic area. In the Third Generation Partnership Project (3GPP), the term "cell" can refer to a coverage area of a base station **310** and/or a base station subsystem serving the coverage area, depending on the context in which the term is used.

[0056] A base station **310** may provide communication coverage for a macro cell, a pico cell, a femto cell, and/or another type of cell. A macro cell may cover a relatively large geographic area (e.g., several kilometers in radius) and may allow unrestricted access by UEs **320** with service subscriptions. A pico cell may cover a relatively small geographic area and may allow unrestricted access by UEs **320** with service subscription. A femto cell may cover a relatively small geographic area (e.g., a home) and may allow restricted access by UEs **320** having association with the femto cell (e.g., UEs **320** in a closed subscriber group (CSG)). A base station **310** for a macro cell may be referred to as a macro base station. A base station **310** for a pico cell may be referred to as a pico base station. A base station **310** for a femto cell may be referred to as a femto base station or an in-home base station. In the example shown in FIG. **3**, the BS **310***a* may be a macro base station for a macro cell **302***a*, the BS **310***b* may be a pico base station for a pico cell **302***b*, and the BS **310***c* may be a femto base station for a femto cell **302***c*. A base station may support one or multiple (e.g., three) cells.

[0057] In some examples, a cell may not necessarily be stationary, and the geographic area of the cell may move according to the location of a base station **310** that is mobile (e.g., a mobile base station). In some examples, the base stations **310** may be interconnected to one another and/or to one or more other base stations **310** or network nodes (not shown) in the wireless network **300** through various types of backhaul interfaces, such as a direct physical connection or a virtual network, using any suitable transport network.

[0058] The wireless network **300** may include one or more relay stations. A relay station is an entity that can receive a transmission of data from an upstream station (e.g., a base station **310** or a UE **320**) and send a transmission of the data to a downstream station (e.g., a UE **320** or a base station **310**). A relay station may be a UE **320** that can relay transmissions for other UEs **320**. In the example shown in FIG. **3**, the BS **310***d* (e.g., a relay base station) may communicate with the BS **310***a* (e.g., a macro base station) and the UE **320***d* in order to facilitate communication between the

BS **310**a and the UE **320**d. A base station **310** that relays communications may be referred to as a relay station, a relay base station, a relay, or the like.

[0059] The wireless network **300** may be a heterogeneous network that includes base stations **310** of different types, such as macro base stations, pico base stations, femto base stations, relay base stations, or the like. These different types of base stations **310** may have different transmit power levels, different coverage areas, and/or different impacts on interference in the wireless network **300**. For example, macro base stations may have a high transmit power level (e.g., 5 to 40 watts) whereas pico base stations, femto base stations, and relay base stations may have lower transmit power levels (e.g., 0.1 to 2 watts).

[0060] A network controller **330** may couple to or communicate with a set of base stations **310** and may provide coordination and control for these base stations **310**. The network controller **330** may communicate with the base stations **310** via a backhaul communication link. The base stations **310** may communicate with one another directly or indirectly via a wireless or wireline backhaul communication link. For example, in some aspects, the wireless network **300** may be, include, or be included in a wireless backhaul network, sometimes referred to as an IAB network. In an IAB network, at least one base station (e.g., base station **310**) may be an anchor base station that communicates with a core network via a wired backhaul link, such as a fiber connection. An anchor base station may also be referred to as an IAB donor (or IAB-donor), a central entity, a central unit, and/or the like. An IAB network may include one or more non-anchor base stations, sometimes referred to as relay base stations or IAB nodes (or IAB-nodes). The non-anchor base station may communicate directly with or indirectly with (e.g., via one or more non-anchor base stations) the anchor base station via one or more backhaul links to form a backhaul path to the core network for carrying backhaul traffic. Backhaul links may be wireless links. Anchor base station(s) and/or non-anchor base station(s) may communicate with one or more UEs (e.g., UE **320**) via access links, which may be wireless links for carrying access traffic.

[0061] In some aspects, a radio access network that includes an IAB network may utilize millimeter wave technology and/or directional communications (e.g., beamforming, precoding and/or the like) for communications between base stations and/or UEs (e.g., between two base stations, between two UEs, and/or between a base station and a UE). For example, wireless backhaul links between base stations may use millimeter waves to carry information and/or may be directed toward a target base station using beamforming, precoding, and/or the like. Similarly, wireless access links between a UE and a base station may use millimeter waves and/or may be directed toward a target wireless node (e.g., a UE and/or a base station). In this way, inter-link interference may be reduced.

[0062] An IAB network may include an IAB donor that connects to a core network via a wired connection (e.g., a wireline backhaul). For example, an Ng interface of an IAB donor may terminate at a core network. Additionally, or alternatively, an IAB donor may connect to one or more devices of the core network that provide a core access and mobility management function (AMF). In some aspects, an IAB donor may include a base station **310**, such as an anchor base station. An IAB donor may include a central unit, which may perform access node controller (ANC) functions and/or AMF functions. The central unit may configure a DU of the IAB donor and/or may configure one or more IAB nodes (e.g., a mobile termination (MT) function and/or a DU function of an IAB node) that connect to the core network via the IAB donor. Thus, a central unit of an IAB donor may control and/or configure the entire IAB network (or a portion thereof) that connects to the core network via the IAB donor, such as by using control messages and/or configuration messages (e.g., a radio resource control (RRC) configuration message or an F1 application protocol (F1AP) message).

[0063] The MT functions of an IAB node (e.g., a child node) may be controlled and/or scheduled by another IAB node (e.g., a parent node of the child node) and/or by an IAB donor. The DU functions of an IAB node (e.g., a parent node) may control and/or schedule other IAB nodes (e.g., child nodes of the parent node) and/or UEs **320**. Thus, a DU may be referred to as a scheduling node or a scheduling component, and an MT may be referred to as a scheduled node or a scheduled component. In some aspects, an IAB donor may include DU functions and not MT functions. That is, an IAB donor may configure, control, and/or schedule communications of IAB nodes and/or UEs **320**. A UE **320** may include only MT functions, and not DU functions. That is, communications of a UE **320** may be controlled and/or scheduled by an IAB donor and/or an IAB node (e.g., a parent node of the UE **320**).

[0064] When a first node controls and/or schedules communications for a second node (e.g., when the first node provides DU functions for the second node's MT functions), the first node may be referred to as a parent node of the second node, and the second node may be referred to as a child node of the first node. A child node of the second node may be referred to as a grandchild node of the first node. Thus, a DU function of a parent node may control and/or schedule communications for child nodes of the parent node. A parent node may be an IAB donor or an IAB node, and a child node may be an IAB node or a UE **320**. Communications of an MT function of a child node may be controlled and/or scheduled by a parent node of the child node.

[0065] A link between a UE **320** and an IAB donor, or between a UE **320** and an IAB node, may be referred to as an access link. An access link may be a wireless access link that provides a UE **320** with radio access to a core network via an IAB donor, and optionally via one or more IAB nodes. Thus, the wireless network **300** may be referred to as a multi-hop network or a wireless multi-hop network.

[0066] A link between an IAB donor and an IAB node or between two IAB nodes may be referred to as a backhaul link. A backhaul link may be a wireless backhaul link that provides an IAB node with radio access to a core network via an IAB donor, and optionally via one or more other IAB nodes. In an IAB network, network resources for wireless communications (e.g., time resources, frequency resources, and/or spatial resources) may be shared between access links and backhaul links. In some aspects, a backhaul link may be a primary backhaul link or a secondary backhaul link (e.g., a backup backhaul link). In some aspects, a secondary backhaul link may be used if a primary backhaul link fails, becomes congested, and/or becomes overloaded, among other examples.

[0067] The UEs **320** may be dispersed throughout the wireless network **300**, and each UE **320** may be stationary or mobile. A UE **320** may include, for example, an access terminal, a terminal, a mobile station, and/or a subscriber unit. A UE **320** may be a cellular phone (e.g., a smart phone), a personal digital assistant (PDA), a wireless modem, a wireless communication device, a handheld device, a laptop computer, a cordless phone, a wireless local loop (WLL) station, a tablet, a camera, a gaming device, a netbook, a smartbook, an ultrabook, a medical device, a biometric device, a wearable device (e.g., a smart watch, smart clothing, smart glasses, a smart wristband, smart jewelry (e.g., a smart ring or a smart bracelet)), an entertainment device (e.g., a music device, a video device, and/or a satellite radio), a vehicular component or sensor, a smart meter/sensor, industrial manufacturing equipment, a global positioning system device, and/or any other suitable device that is configured to communicate via a wireless medium.

[0068] Some UEs **320** may be considered machine-type communication (MTC) or evolved or enhanced machine-type communication (eMTC) UEs. An MTC UE and/or an eMTC UE may include, for example, a robot, a drone, a remote device, a sensor, a meter, a monitor, and/or a location tag, that may communicate with a base station, another device (e.g., a remote device), or some other entity. Some UEs **320** may be considered Internet-of-Things (IoT) devices, and/or may be implemented as NB-IoT (narrowband IoT) devices. Some UEs **320** may be considered a customer premises equipment. A UE **320** may be included inside a housing that houses components of the UE **320**, such as processor components and/or memory components. In some examples, the processor components and the memory components may be coupled together. For example, the processor components (e.g., one or more processors) and the memory components (e.g., a memory) may be operatively coupled, communicatively coupled, electronically coupled, and/or electrically coupled.

[0069] In general, any number of wireless networks **300** may be deployed in a given geographic area. Each wireless network **300** may support a particular RAT and may operate on one or more frequencies. A RAT may be referred to as a radio technology, an air interface, or the like. A frequency may be referred to as a carrier, a frequency channel, or the like. Each frequency may support a single RAT in a given geographic area in order to avoid interference between wireless networks of different RATs. In some cases, NR or 5G RAT networks may be deployed.

[0070] In some examples, two or more UEs **320** (e.g., shown as UE **320***a* and UE **320***e*) may communicate directly using one or more sidelink channels (e.g., without using a base station **310** as an intermediary to communicate with one another). For example, the UEs **320** may communicate using peer-to-peer (P2P) communications, device-to-device (D2D) communications, a vehicle-to-everything (V2X) protocol (e.g., which may include a vehicle-to-vehicle (V2V) protocol, a vehicle-to-infrastructure (V2I) protocol, or a vehicle-to-pedestrian (V2P) protocol), and/or a mesh network. In such examples, a UE **320** may perform scheduling operations, resource selection operations, and/or other operations described elsewhere herein as being performed by the base station **310**.

[0071] Devices of the wireless network **300** may communicate using the electromagnetic spectrum, which may be subdivided by frequency or wavelength into various classes, bands, channels, or the like. For example, devices of the wireless network **300** may communicate using one or more operating bands. In 5G NR, two initial operating bands have been identified as frequency range designations FR1 (410 MHz-7.125 GHz) and FR2 (24.25 GHz-52.6 GHz). It should be understood that although a portion of FR1 is greater than 6 GHz, FR1 is often referred to (interchangeably) as a “Sub-6 GHz” band in various documents and articles. A similar nomenclature issue sometimes occurs with regard to FR2, which is often referred to (interchangeably) as a “millimeter wave” band in documents and articles, despite being different from the extremely high frequency (EHF) band (30 GHz-300 GHz) which is identified by the International Telecommunications Union (ITU) as a “millimeter wave” band.

[0072] The frequencies between FR1 and FR2 are often referred to as mid-band frequencies. Recent 5G NR studies have identified an operating band for these mid-band frequencies as frequency range designation FR3 (7.125 GHz-24.25 GHz). Frequency bands falling within FR3 may inherit FR1 characteristics and/or FR2 characteristics, and thus may effectively extend features of FR1 and/or FR2 into mid-band frequencies. In addition, higher frequency bands are currently being explored to extend 5G NR operation beyond 52.6 GHz. For example, three higher operating bands have been identified as frequency range designations FR4a or FR4-1 (52.6 GHz-71 GHz), FR4 (52.6 GHz-114.25 GHz), and FR5 (114.25 GHz-300 GHz). Each of these higher frequency bands falls within the EHF band.

[0073] With the above examples in mind, unless specifically stated otherwise, it should be understood that the term “sub-6 GHz” or the like, if used herein, may broadly represent frequencies that may be less than 6 GHz, may be within FR1, or may include mid-band frequencies. Further, unless specifically stated otherwise, it should be understood that the term “millimeter wave” or the like, if used herein, may broadly represent frequencies that may include mid-band frequencies, may be within FR2, FR4, FR4-a or FR4-1, and/or FR5, or may be within the EHF band. It is contemplated that the frequencies included in these operating bands (e.g., FR1, FR2, FR3, FR4, FR4-a, FR4-1, and/or FR5) may be modified, and techniques described herein are applicable to those modified frequency ranges.

[0074] As described above, in some aspects, a network node (e.g., the network node **102**, **104**, and/or the network node **106** depicted in FIG. **1**) may be implemented in a wireless communication environment. For example, in some aspects, the network node may be implemented as a UE (e.g., UE **320***a*) a base station (e.g., base station **310***a*), relay device, and/or TRP, among other examples. In some such aspects, as shown in FIG. **3**, the UE **320***a* may include a communication manager **340** and/or a transceiver **345** and the base station **310***a* may include a communication manager **350** and/or a transceiver **355**. In some aspects, the communication manager **340** and/or **350** may be, be similar to, include, or be included in, the communication manager **110** and/or **114** depicted in FIG. **1** and/or the communication manager **235** depicted in FIG. **2**. In some aspects, the transceiver **345** and/or **355** may be, be similar to, include, or be included in, the transceiver **112** and/or **116** depicted in FIG. **1**. In some aspects, the transceiver **345** and/or **355** may include, or be included in, the communication interface **230** depicted in FIG. **2**.

[0075] As indicated above, FIG. **3** is provided as an example. Other examples may differ from what is described with regard to FIG. **3**.

[0076] FIG. **4** is a diagram illustrating an environment **400** including a network node **402** in wireless communication with a network node **404** (e.g., via a network such as the network **108** depicted in FIG. **1** and/or the wireless network **300** depicted in FIG. **3**), in accordance with the present disclosure. The network node **402** may be equipped with a set of antennas **406***a* through **406***t*, such as T antennas (T≥1). The network node **404** may be equipped with a set of antennas **408***a* through **408***r*, such as R antennas (R≥1).

[0077] At the network node **402**, a transmit processor **410** may receive data, from a data source **412**, intended for the network node **404** (or a set of network nodes **404**). The transmit processor **410** may select one or more modulation and coding schemes (MCSs) for the network node **404** based on one or more channel quality indicators (CQIs) received from that network node **404**. The network node **402** may process (e.g., encode and modulate) the data for the network node **404** based on the MCS(s) selected for the network node **404** and may provide data symbols for the network node **404**. The transmit processor **410** may process system information (e.g., for semi-static resource partitioning information (SRPI)) and control information (e.g., CQI requests, grants, and/or upper layer signaling) and provide overhead symbols and control symbols. The transmit processor **410** may generate reference symbols for reference signals (e.g., a cell-specific reference signal (CRS) or a demodulation reference signal (DMRS)) and synchronization signals (e.g., a primary synchronization signal (PSS) or a secondary synchronization signal (SSS)). A transmit (TX) multiple-input multiple-output (MIMO) processor **414** may perform spatial processing (e.g., precoding) on the data symbols, the control symbols, the overhead symbols, and/or the reference symbols, if applicable, and may provide a set of output symbol streams (e.g., T output symbol streams) to a corresponding set of modems **416***a* through **416***t* (e.g., T modems). For example, each output symbol stream may be provided to a modulator component (shown as MOD) of a modem of the set of modems **416***a* through **416***t*. Each modem of the set of modems **416***a* through **416***t* may use a respective modulator component to process a respective output symbol stream (e.g., for OFDM) to obtain an output sample stream. Each modem of the set of modems **416***a* through **416***t* may further use a respective modulator component to process (e.g., convert to analog, amplify, filter, and/or upconvert) the output sample stream to obtain a signal. One or more modems of the set of modems **416***a* through **416***t* may transmit a set of signals (e.g., T signals) via a corresponding antenna of the set of antennas **406***a* through **406***t*. The signal may include, for example, a downlink signal.

[0078] At the network node **404**, one or more antennas of the set of antennas **408***a* through **408***r* may receive the signals from the network node **402** and/or network nodes and may provide a set of received signals (e.g., R received signals) to one or more modems of a set of modems **418***a* through **418***r* (e.g., R modems). For example, each received signal may be provided to a demodulator component (shown as DEMOD) of a respective modem of the set of modems **418***a* through **418***r*. Each modem of the set of modems **418***a* through **418***r* may use a respective demodulator component to condition (e.g., filter, amplify, downconvert, and/or digitize) a received signal to obtain input samples. Each modem of the set of modems **418***a* through **418***r* may use a demodulator component to further process the input samples (e.g., for OFDM) to obtain received symbols. A MIMO detector **420** may obtain received symbols from one or more modems of the set of modems **418***a* through **418***r*, may perform MIMO detection on the received symbols if applicable, and may provide detected symbols.

[0079] A receive processor **422** may process (e.g., demodulate and decode) the detected symbols, may provide decoded data for the network node **404** to a data sink **424**, and may provide decoded control information and system information to a controller/processor **426**. The term "controller/processor" may refer to one or more controllers, one or more processors, or a combination thereof. The controller/processor **426** may be, be similar to, include, or be included in, the processor **210** depicted in FIG. **2**. The controller/processor **426** may determine a reference signal received power (RSRP) parameter, a received signal strength indicator (RSSI) parameter, a reference signal received quality (RSRQ) parameter, and/or a CQI parameter, among other examples.

[0080] A network controller **428** may include a communication unit **430**, a controller/processor **432**, and a memory **434**. The network controller **428** may be, be similar to, include, or be included in, the network controller **330** depicted in FIG. **3**. The network controller **428** may include, for example, one or more devices in a core network. The network controller **428** may communicate with the network node **402** via the communication unit **430**.

[0081] One or more antennas (e.g., antennas **406***a* through **406***t* and/or antennas **408***a* through **408***r*) may include, or may be included within, one or more antenna panels, one or more antenna groups, one or more sets of antenna elements, and/or one or more antenna arrays, among other examples. An antenna panel, an antenna group, a set of antenna elements, and/or an antenna array may include one or more antenna elements (within a single housing or multiple housings), a set of coplanar antenna elements, a set of non-coplanar antenna elements, and/or one or more antenna elements coupled to one or more transmission and/or reception components, such as one or more components of FIG. **4**.

[0082] Similarly, at the network node **404**, a transmit processor **436** may receive and process data from a data source **438** and control information (e.g., for reports that include RSRP, RSSI, RSRQ, and/or CQI) from the controller/processor **426**. The transmit processor **436** may generate reference symbols for one or more reference signals. The symbols from the transmit processor **436** may be precoded by a TX MIMO processor **440** if applicable, and further processed by one or more of the set of modems **418***a* through **418***r* (e.g., for DFT-s-OFDM or CP-OFDM), and transmitted to the network node **402**. In some examples, each modem of the set of modems **418***a* through **418***r* of the network node **404** may include a modulator and a demodulator. In some examples, the network node **404** includes a transceiver. The transceiver may include any combination of the antenna(s) **408***a* through **408***r*, the modem(s) **418***a* through **418***r*, the MIMO detector **420**, the receive processor **422**, the transmit processor **436**, and/or the TX MIMO processor **440**. The transceiver may be, be similar to, include, or be included in, the communication interface **230** depicted in FIG. **2**. The transceiver may be used by a processor (e.g., the controller/processor **426**) and/or a memory **442** to perform aspects of any of the methods described herein (e.g., with reference to FIGS. **7-11**).

[0083] At the network node **402**, the signals from network node **404** and/or other network nodes may be received by one or more antennas of the set of antennas **406***a* through **406***t*, processed by one or more modems of the set of modems **416***a* through **416***t* (e.g., a demodulator component, shown as DEMOD), detected by a MIMO detector **444** if applicable, and further processed by a receive processor **446** to obtain decoded data and control information sent by the network node **404**. The receive processor **446** may provide the decoded data to a data sink **448** and provide the decoded control information to a controller/processor **450**. The network node **402** may include a communication unit **452** and may communicate with the network controller **428** via the communication unit **452**. The network node **402** may include a scheduler **454** to schedule one or more network nodes **404** for downlink and/or uplink communications. In some examples, one or more modems of the set of modem **416***a* through **416***t* of the network node **402** may include a modulator and a demodulator. In some examples, the network node **402** includes a transceiver. The transceiver may include any combination of the antenna(s) **406***a* through **406***t*, the modem(s) **416***a* through **416***t*, the MIMO detector **444**, the receive processor **446**, the transmit processor **410**, and/or the TX MIMO processor **414**. The transceiver may be, be similar to, include, or be included in, the communication interface **230** depicted in FIG. **2**. The transceiver may be used by a processor (e.g., the controller/processor **450**) and a memory **456** to perform aspects of any of the methods described herein (e.g., with reference to FIGS. **7-11**).

[0084] The controller/processor **450** of the network node **402**, the controller/processor **426** of the network node **404**, and/or any other component(s) of FIG. **4** may perform one or more techniques associated with reporting CLI associated with an electromagnetic radiation reflection relay service, as described in more detail elsewhere herein. For example, the controller/processor **450** of the network node **402**, the controller/processor **426** of the network node **404**, and/or any other component(s) of FIG. **4** may perform or direct operations of, for example, process **800** of FIG. **8**, process **900** of FIG. **9**, process **1000** of FIG. **10**, and/or other processes as described herein. The memory **442** and the memory **456** may store data and program codes for the network node **402** and the network node **404**, respectively. In some examples, the memory **442** and/or the memory **456** may include a non-transitory computer-readable medium storing one or more instructions (e.g., code and/or program code) for wireless communication. For example, the one or more instructions, when executed (e.g., directly, or after compiling, converting, and/or interpreting) by one or more respective processors of the network node **402** and/or the network node **404**, may cause the one or more processors, the network node **404**, and/or the network node **402** to perform or direct operations of, for example, process **800** of FIG. **8**, process **900** of FIG. **9**, process **1000** of FIG. **10**, and/or other processes as described herein. In some examples, executing instructions may include running the instructions, converting the instructions, compiling the instructions, and/or interpreting the instructions, among other examples.

[0085] In some aspects, a first network node includes means for receiving a signal comprising a first signal characteristic associated with a first pre-coding vector corresponding to a first communication channel between the first network node and a second network node and a second signal characteristic associated with a second pre-coding vector corresponding to a second communication channel between the first network node and an electromagnetic radiation reflection relay network node; and/or means for transmitting channel estimation information corresponding to the signal, wherein the channel estimation information includes first channel estimation information and second channel estimation information, wherein the first channel estimation information corresponds to the first communication channel and is based on the first pre-coding vector, and the second channel estimation information corresponds to the second communication channel and is based on a second pre-coding vector associated with the second communication channel.

[0086] In some aspects, the first network node includes means for transmitting CLI configuration information that includes first information indicative of a reference signal resource; means for receiving channel estimation information corresponding to a signal that is based on the reference signal resource, wherein the channel estimation information includes first channel estimation information and second channel estimation information, wherein the first channel estimation information corresponds to a first communication channel between a second network node and a third network node and is based on a first pre-coding vector associated with the first communication channel, and the second channel estimation information corresponds to a second communication channel between an electromagnetic radiation reflection relay network node and the third network node and is based on a second pre-coding vector associated with the second communication channel; and/or means for transmitting reflection relay configuration information to the electromagnetic radiation reflection relay network node, wherein the reflection relay configuration information is indicative of one or more electromagnetic reflection parameters based on the channel estimation information.

[0087] In some aspects, the first network node includes means for receiving, from an aggressor network node, a plurality of aggressor reference signals, wherein the plurality of aggressor reference signals includes a first aggressor reference signal pre-coded by first pre-coding information and a second aggressor reference signal pre-coded by second pre-coding information; means for receiving, from an electromagnetic radiation reflection relay network node, a plurality of reflected reference signals, wherein the plurality of reflected reference signals includes a first reflected reference signal pre-coded by third pre-coding information and a second reflected reference signal pre-coded by fourth pre-coding information, wherein the first aggressor reference signal and the first reflected reference signal overlap in time, and the second aggressor reference signal and the second reflected reference signal overlap in time; and/or means for transmitting, to a second network node, channel estimation information including first channel estimation information and second channel estimation information, wherein the channel estimation information is based on the plurality of aggressor reference signals and the plurality of reflected reference signals, wherein the first channel information is indicative of one or more channel characteristics corresponding to a first channel between the first network node and the aggressor network node, and wherein the second channel information is indicative of one or more channel characteristics corresponding to a second channel between the first network node and the electromagnetic radiation reflection relay network node. In some aspects, the means for the network node to perform operations described herein may include, for example, one or more of communication manager **458** or **460**, transmit processor **410** or **436**, TX MIMO processor **414** or **440**, modem **416***a*-**416***t* or **418***a*-**418***r*, antenna **406***a*-**406***t* or **408***a*-**408***r*, MIMO detector **420** or **444**, receive processor **422** or **446**, controller/processor **426** or **450**, memory **442** or **456**, or scheduler **454**.
[0088] While blocks in FIG. **4** are illustrated as distinct components, the functions described above with respect to the blocks may be implemented in a single hardware, software, or combination component or in various combinations of components. For example, the functions described with respect to the transmit processor **436**, the receive processor **422**, and/or the TX MIMO processor **440** may be performed by or under the control of the controller/processor **426**. Any number of other combination of various combinations of components depicted in FIG. **4** may be within the ambit of the present disclosure.
[0089] As indicated above, FIG. **4** is provided as an example. Other examples may differ from what is described with regard to FIG. **4**.
[0090] FIG. **5** is a diagram illustrating an example **500** of an O-RAN architecture, in accordance with the present disclosure. As shown in FIG. **5**, the O-RAN architecture may include a CU **510** that communicates with a core network **520** via a backhaul link. Furthermore, the CU **510** may communicate with one or more DUs **530** via respective midhaul links. The DUs **530** may each communicate with one or more RUs **540** via respective fronthaul links, and the RUs **540** may each communicate with respective UEs **320** via RF access links. The DUs **530** and the RUs **540** may also be referred to as O-RAN DUS (O-DUs) **530** and O-RAN RUs (O-RUs) **540**, respectively.
[0091] In some aspects, the DUs **530** and the RUs **540** may be implemented according to a functional split architecture in which functionality of a base station **310** (e.g., an eNB or a gNB) is provided by a DU **530** and one or more RUs **540** that communicate over a fronthaul link. Accordingly, as described herein, a base station **310** may include a DU **530** and one or more RUs **540** that may be co-located or geographically distributed. In some aspects, the DU **530** and the associated RU(s) **540** may communicate via a fronthaul link to exchange real-time control plane information via a lower layer split (LLS) control plane (LLS-C) interface, to exchange non-real-time management information via an LLS management plane (LLS-M) interface, and/or to exchange user plane information via an LLS user plane (LLS-U) interface.
[0092] Accordingly, the DU **530** may correspond to a logical unit that includes one or more base station functions to control the operation of one or more RUs **540**. For example, in some aspects,

the DU **530** may host a radio link control (RLC) layer, a medium access control (MAC) layer, and one or more high physical (PHY) layers (e.g., forward error correction (FEC) encoding and decoding, scrambling, and/or modulation and demodulation) based on a lower layer functional split. Higher layer control functions, such as a packet data convergence protocol (PDCP), RRC, and/or service data adaptation protocol (SDAP), may be hosted by the CU **510**. The RU(s) **540** controlled by a DU **530** may correspond to logical nodes that host RF processing functions and low-PHY layer functions (e.g., fast Fourier transform (FFT), inverse FFT (iFFT), digital beamforming, and/or physical random access channel (PRACH) extraction and filtering) based on the lower layer functional split. Accordingly, in an O-RAN architecture, the RU(s) **540** handle all over the air (OTA) communication with a UE **320**, and real-time and non-real-time aspects of control and user plane communication with the RU(s) **540** are controlled by the corresponding DU **530**, which enables the DU(s) **530** and the CU **510** to be implemented in a cloud-based RAN architecture.

[0093] As indicated above, FIG. **5** is provided as an example. Other examples may differ from what is described with regard to FIG. **5**.

[0094] FIG. **6** is a diagram illustrating an example **600** of wireless communication, in accordance with the present disclosure. As shown, a first network node **602** and a second network node **604** can communicate via a communication channel **606**. An electromagnetic radiation reflective relay network node **608** (shown as "reflective network node") can be used to relay signals from the network node **602** to the network node **604** via a communication channel **610**. The electromagnetic radiation reflective relay network node **608** can include, for example, a radio frequency reflection array configured to perform radio frequency reflection services. The electromagnetic radiation reflective relay network node **608** can be, for example, a reconfigurable intelligent surface (RIS) (which also can be referred to as an intelligent reflective surface (IRS)).

[0095] As shown, for example, the network node **602** can transmit a signal, via a communication channel **612**, to the electromagnetic radiation reflective relay network node **608**, which can relay the signal, via the communication channel **610**, to the network node **604**. In some cases, the electromagnetic radiation reflective relay network node **608** can relay transmissions between the network node **602** and a network node **614**. For example, as shown, the network node **602** can transmit, via the communication channel **612**, a signal to the electromagnetic radiation reflective relay network node **608**, which can relay the signal, via a communication channel **616**, to the network node **614**.

[0096] As shown in FIG. **6**, the electromagnetic radiation reflective relay network node **608** may include a set of reflecting elements **618** disposed adjacent to a ground plane **620**. Each reflecting element **618** can be coupled to a phase shifting component **622**, and each phase shifting component **622** can be coupled to a respective grounding component **624**. In some aspects, each reflecting element **618** can be coupled to two phase shifting components **622**, one for each polarization. In some aspects, one or more reflecting elements **618** can be driven by a power amplifier **626**. The power amplifier **626** can be coupled to a power supply **628** and can be controlled by a controller **630**. In some cases, for example, the power amplifier **626** can be configured to provide just enough power to offset energy loss due to reflection of a signal and/or phase adjustment thereof. In some cases, a complexity of the controller **630** and/or power consumption by the power amplifier **626** can be based on selection of phase shifting components **622**. In some cases, the electromagnetic radiation reflective relay network node **608** can be configured to reflect a signal **632** (e.g., the signal transmitted via the communication channel **612**) by beamforming a reflected signal **634** (e.g., the signal transmitted via the communication channel **616**) to direct the reflected signal **634** based on one or more beams **636**.

[0097] In some cases, as shown, the network node **602** can transmit, via a communication channel **638**, a signal to the network node **614** without using the electromagnetic radiation reflective relay network node **608** as a relay. The network node **614** can transmit, via a communication channel

640, a signal to the network node **604**. In some cases, the network node **614** can transmit, via a communication channel **642**, a signal to the electromagnetic radiation reflective relay network node **608**, which can relay, via the communication channel **610**, the signal to the network node **604**. In some cases, the network node **614** can transmit, via a communication channel **644**, configuration information to the electromagnetic radiation reflective relay network node **608** to configure one or more aspects of the operation of the electromagnetic radiation reflective relay network node **608**.

[0098] In a communication scenario such as the scenario depicted in FIG. **6**, CLI may impede efficiency and accuracy of communications. CLI is the interference from one entity to another nearby entity. For example, CLI can occur between two network nodes, such as UEs, when a network configures different time domain duplexing (TDD) UL and DL slot formats to nearby UEs. When a network node (e.g., network node **602**) (which may be referred to as an aggressor network node) is transmitting, another network node (e.g., network node **604**), (which may be referred to as a victim network node) may receive this transmission as CLI in its DL symbols if the aggressor network node's UL symbol collide with at least one DL symbol of the victim network node. As shown in FIG. **6**, the aggressor network node **602** can communicate with the network node **614** via the channel **638** (which can be indicated as “h.sub.A-g”). The aggressor network node **602** can interfere with the victim network node **604** via CLI in the channel **606** (which may be indicated as “h.sub.A-V”). In some cases, the electromagnetic radiation reflective relay network node **608** can be configured to maximize a channel strength of the channel **616** (which can be indicated as “h.sub.A-I-g”). The aggressor network node **602** also may interfere with the victim network node **604** via CLI in the reflected channel **610** (which may be indicated as “h.sub.A-I-V”). In some cases, the victim network node **604** can measure CLI from the aggressor network node **602**. For example, the victim network node **604** can measure CLI if the network configures one or more CLI measurement resources to do so. CLI measurement is defined as periodic measurement based on sounding reference signal (SRS) RSRP or RSSI. The CLI can reduce the effectiveness of communications, thereby negatively impacting network performance.

[0099] Some techniques and apparatuses described herein provide for measurement of CLI associated with an electromagnetic radiation reflective relay network node and mitigation of CLI based on those measurements. For example, in some aspects, the network node **604** may receive a signal (which may be indicated as “y.sub.T1”) that includes a first signal characteristic (which may be indicated as “h.sub.A-Vc.sub.11s”) associated with a first pre-coding vector, c.sub.11, corresponding to a first communication channel, h.sub.A-V, (e.g., the communication channel **606**) between the network node **604** and the network node **602** and a second signal characteristic (which may be indicated as h.sub.A-I-Vc.sub.12s”) associated with a second pre-coding vector, c.sub.12, corresponding to a second communication channel (e.g., the communication channel **610**) between the network node **604** and the electromagnetic radiation reflection relay network node. The network node **604** may transmit, and the network node **614** may receive, channel estimation information corresponding to the signal. The channel estimation information may include first channel estimation information and second channel estimation information. The first channel estimation information corresponds to the first communication channel and is based on the first pre-coding vector, and the second channel estimation information corresponds to the second communication channel and is based on a second pre-coding vector.

[0100] In some aspects, the network node **614** may transmit reflection relay configuration information to the electromagnetic radiation reflection relay network node **608**. The reflection relay configuration information may be indicative of one or more electromagnetic reflection parameters based on the channel estimation information. In some aspects, the electromagnetic radiation reflection relay network node **608** may be used to manipulate and/or add a signal propagation path for mitigating CLI impact. For example, in some aspects, reflection relay configuration information may be used to configure the electromagnetic radiation reflection relay network node **608** to increase, or maximize, a signal strength of signals transmitted via the communication channel **616**, h.sub.A-I-g. In some aspects, the reflection relay configuration information may be used to configure the electromagnetic radiation reflection relay network node **608** to reduce, or minimize, a signal strength of signals transmitted via the communication channel **616**, h.sub.A-I-g. In some aspects, the reflection relay configuration information may be used to configure the electromagnetic radiation reflection relay network node **608** to reduce, or minimize, CLI in the communication channel **606**. For example, CLI in the communication channel **610** may nullify (e.g., cancel) all or part of the CLI in the communication channel **606**. In this way, aspects described herein may provide for utilizing joint estimation of two channels based on one or more resources configured for CLI measurement associated with an electromagnetic radiation reflection relay network node to facilitate mitigation of CLI, thereby positively impacting network performance.

[0101] As indicated above, FIG. **6** is provided as an example. Other examples may differ from what is described with regard to FIG. **6**.

[0102] FIG. **7** is a diagram illustrating an example **700** associated with nonlinear modeling for channel estimation, in accordance with the present disclosure. As shown, a network node **702**, a network node **704**, a network node **706**, and a network node **708** may communicate with one another. The network nodes **702**, **704**, **706**, and/or **708** may be, or be similar, to the network node **102** and/or the network node **104** depicted in FIG. **1**. The network node **702** may be, or be similar to, the network node **614** depicted in FIG. **6**. The network node **704** may be a victim network node and may be, or be similar to, the network node **604** depicted in FIG. **6**. The network node **706** may be an aggressor network node and may be, or be similar to, the network node **602** depicted in FIG. **6**. The network node **708** may be an electromagnetic radiation reflection relay network node and may be, or be similar to, the electromagnetic radiation reflection relay network node **608** depicted in FIG. **6**.

[0103] As shown by reference number **710**, the first network node **702**, the second network node **704**, the third network node **706**, and/or the fourth network node **708** may perform an aggressor network node identification process to identify an aggressor network node. For example, in some aspects, the network node **702** may receive information from one or more of the network nodes **704**, **706**, and/or **708**. The network node **702** may determine, based on the received information, that the network node **706** is an aggressor network node. In some aspects, the network node **704** may determine that the network node **702** is an aggressor network node.

[0104] As shown by reference number **712**, the network node **702** may transmit, and the network node **704**, may receive, CLI configuration information. Similarly, as shown by reference number **714**, the network node **702** may transmit, and the network node **706** may receive, CLI configuration information. In some aspects, the CLI configuration information transmitted to the network node **704** may be the same as the CLI configuration information transmitted to the network node **706**. In some other aspects, the CLI configuration information transmitted to the network node **704** may be at least partially different than the CLI configuration information transmitted to the network node **706**. For example, the CLI configuration information may configure the network node **704** with CLI measurement resources and/or CLI measurement parameters that may be used by the network node **704** to measure CLI. The CLI configuration information may configure the network node **706** with pre-coding information, CLI reference signal resources (e.g., SRS resources), and/or CLI reference signal transmission parameters.

[0105] In some aspects, the CLI configuration information transmitted to the network node **704** may specify the network node **706** and the network node **708** (e.g., the electromagnetic radiation reflection relay network node). In some aspects, the CLI configuration information transmitted to the network node **704** and/or the network node **706** may include first information indicative of a reference signal resource. In some aspects, the CLI configuration information transmitted to the network node **704** and/or **706** may include second information indicative of a plurality of pre-coding vectors corresponding to a pre-coding matrix. The plurality of pre-coding vectors may include the first pre-coding vector and the second pre-coding vector. In some aspects, the plurality of pre-coding vectors may include at least one additional pre-coding vector associated with at least one additional network node configured to provide at least one additional electromagnetic radiation reflection relay service. In some aspects, the network node **704** may include a memory that includes non-signaled information stored thereon, where the non-signaled information includes the first pre-coding vector and the second pre-coding vector.

[0106] As shown by reference number **716**, the network node **704** may receive a signal. In some aspects, the signal may include a reference signal such as, for example, an SRS. In some aspects, the signal may include a first signal characteristic **718** associated with a first pre-coding vector corresponding to a first communication channel between the network node **704** and the network node **702**. In some aspects, the first signal characteristic may be referred to as a first "aggressor reference signal," and may be pre-coded by first pre-coding information (e.g., the first pre-coding vector). For example, the first signal characteristic may include a first aggressor reference signal transmitted by the network node **704**, and may be represented as h.sub.A-Vc.sub.11s, where h.sub.A-V is the communication channel between the network node **704** and the network node **702**, c.sub.11 is first pre-coding information (e.g., a first pre-coding vector) associated with the channel h.sub.A-V, and s is a configured CLI resource (e.g., an SRS resource). In some aspects, the network node **704** may receive at least one additional signal. The at least one additional signal may include at least one additional signal characteristic associated with the first communication channel. For example, the at least one additional signal characteristic may include a second aggressor reference signal transmitted by the network node **704**, and may be represented as h.sub.A-Vc.sub.21s, where h.sub.A-V is the first communication channel, between the network node **704** and the network node **702**, c.sub.21 is second pre-coding information (e.g., a pre-coding vector) associated with the channel h.sub.A-V, and s is a configured CLI resource (e.g., an SRS resource) associated with the second aggressor reference signal.

[0107] The signal also may include a second signal characteristic **720** associated with a second pre-coding vector corresponding to a second communication channel between the network node **704** and the network node **706**. The second signal characteristic may be referred to as a "first reflected reference signal," and may be pre-coded by third pre-coding information (e.g., a third pre-coding vector). For example, the first reflected reference signal may be represented as h.sub.A-I-Vc.sub.12s, where h.sub.A-I-V is the second communication channel, between the network node **704** and the network node **706**, c.sub.12 is third pre-coding information, and s is a configured CLI resource associated with the first reflected reference signal. The at least one additional signal may include a fourth reflected reference signal, transmitted from the network node **708** (e.g., as a reflection of a transmission from the network node **702**). The fourth reflected reference signal may be pre-coded using fourth pre-coding information c.sub.22 and may be represented by h.sub.A-I-Vc.sub.22s. In some aspects, the first aggressor reference signal and the first reflected reference signal may overlap in time (e.g., a partial overlap in time or a full overlap in time), and the second aggressor reference signal and the second reflected reference signal may overlap in time (e.g., a partial overlap in time or a full overlap in time).

[0108] Thus, the signal, which may be a first transmission (T.sub.1) of a number of transmissions received by the network node **704** may be represented as y.sub.T1=HC.sub.T1=h.sub.A-Vc.sub.11s+h.sub.A-I-Vc.sub.12s, where H is the communication channel associated with the signal, s is the configured CLI resource, and C.sub.T1={c.sub.11, c.sub.12} is a pre-coding vector associated with the communication channel H at the first transmission T.sub.1. At time T.sub.2, the network node **704** may receive the at least one additional signal y.sub.T2=HC.sub.T2=h.sub.A-Vc.sub.21s+h.sub.A-I-Vc.sub.22s, where the pre-coding vector is defined as C.sub.T2={c.sub.21, c.sub.22}.

[0109] As shown by reference number **722**, the network node **704** may generate channel estimation information. The channel estimation information may include first channel estimation information and second channel estimation information. The first channel estimation information may correspond to the first communication channel and may be based on the first pre-coding vector, and the second channel estimation information may correspond to the second communication channel and may be based on a second pre-coding vector associated with the second communication channel. For example, the channel estimation information may be based on the plurality of aggressor reference signals and the plurality of reflected reference signals. The first channel information may be indicative of one or more channel characteristics corresponding to the first channel, and the second channel information may be indicative of one or more channel characteristics corresponding to the second channel. For example, in sum, Y={y.sub.T1, y.sub.T2}, where Y=HCs, and where, C={C.sub.T1, C.sub.T2} is a 2×2 matrix and can be made known to the network node **704** as a configuration and s is the configured CLI SRS resource. Since C and s may be both known to the network node **704**, H can be estimated as H.sub.est in general via any number of different matrix and/or algebraic math manipulations. For example, in some aspects, H.sub.est=Y*Inverse(Cs).

[0110] As shown by reference number **724**, the network node **704** may transmit, and the network node **702** may receive, the channel estimation information. In some aspects, the channel estimation information may include the first channel estimation information and the second channel estimation information. In some aspects, the channel estimation information may be based on the CLI configuration information. In some aspects, channel estimation may be configured to be reported as per-victim network node based channel estimation information. For example, a network node **704** may be configured (e.g., by the network node **702**) to report channels from an indicated aggressor network node and an indicated electromagnetic radiation reflection relay network node.

[0111] In some aspects, for example, the first channel estimation information may include first raw channel estimation information and the second channel estimation information may include second raw channel estimation information. The first raw channel estimation information and the second raw channel estimation information may be non-differential information and non-quantized information. In some aspects, the first channel estimation information may be indicative of a first channel estimation value associated with the first communication channel, and the second channel estimation information may be a differential value indicative of a second channel estimation value. The differential value may be relative to the first channel estimation value. For example, in some aspects, the differential value may correspond to a ratio between h.sub.A-V-est and h.sub.A-I-V-est. In some aspects, the first channel estimation information may include first quantized channel estimation information, and the second channel estimation information may include second quantized channel estimation information. For example, the channel estimation information may include quantized versions of h.sub.A-V-est and h.sub.A-I-V-est, which may include single respective values indicating the respective channel estimations.

[0112] As shown by reference number **726**, the network node **702** may determine reflection relay configuration information. For example, the reflection relay configuration information may include one or more electromagnetic reflection parameters based on the channel estimation information. In some aspects, the one or more electromagnetic reflection parameters may include at least one of a phase shift coefficient or an amplitude coefficient. In some aspects, the one or more electromagnetic reflection parameters may include one or more electromagnetic reflection parameters determined so as to mitigate CLI associated with at least one of the first communication channel or the second communication channel. For example, in some aspects, the network node **702** may determine the one or more electromagnetic reflection parameters to increase a signal strength associated with the second communication channel. In some aspects, the network node **702** may determine the one or more electromagnetic reflection parameters to decrease a signal strength associated with the second communication channel.

[0113] As shown by reference number **728**, the network node **702** may transmit, and the network node **708** may receive, the reflection relay configuration information. The reflection relay configuration information may be indicative of the one or more electromagnetic reflection parameters based on the channel estimation information.

[0114] As indicated above, FIG. **7** is provided as an example. Other examples may differ from what is described with regard to FIG. **7**. For example, in some aspects, aspects may be applied in a scenario involving two or more electromagnetic radiation reflection relay network nodes. For example, the dimensions of the pre-coding matrix C may be expanded such that the number of columns corresponds to the number of reference signal transmissions and the number of rows is scaled with the number of electromagnetic radiation reflection relay network nodes.

[0115] FIG. **8** is a diagram illustrating an example process **800** performed, for example, by a first network node, in accordance with the present disclosure. Example process **800** is an example where the first network node (e.g., the network node **704**) performs operations associated with reporting CLI associated with an electromagnetic radiation reflection relay service.

[0116] As shown in FIG. **8**, in some aspects, process **800** may include receiving CLI configuration information (block **810**). For example, the first network node (e.g., using communication manager **1108** and/or reception component **1102**, depicted in FIG. **11**) may receive CLI configuration information, as described above.

[0117] As further shown in FIG. **8**, in some aspects, process **800** may include receiving a signal comprising a first signal characteristic associated with a first pre-coding vector corresponding to a first communication channel between the first network node and a second network node and a second signal characteristic associated with a second pre-coding vector corresponding to a second communication channel between the first network node and an electromagnetic radiation reflection relay network node (block **820**). For example, the first network node (e.g., using communication manager **1108** and/or reception component **1102**, depicted in FIG. **11**) may receive a signal comprising a first signal characteristic associated with a first pre-coding vector corresponding to a first communication channel between the first network node and a second network node and a second signal characteristic associated with a second pre-coding vector corresponding to a second communication channel between the first network node and an electromagnetic radiation reflection relay network node, as described above. In some aspects, the signal comprises a reference signal. In some aspects, the reference signal comprises a sounding reference signal.

[0118] As further shown in FIG. **8**, in some aspects, process **800** may include generating channel estimation information corresponding to the signal (block **830**). For example, the first network node (e.g., using communication manager **1108** and/or generation component **1110**, depicted in FIG. **11**) may generate channel estimation information corresponding to the signal, as described above.

[0119] As further shown in FIG. **8**, in some aspects, process **800** may include transmitting the channel estimation information corresponding to the signal, wherein the channel estimation information includes first channel estimation information and second channel estimation information, wherein the first channel estimation information corresponds to the first communication channel and is based on the first pre-coding vector, and the second channel estimation information corresponds to the second communication channel and is based on a second pre-coding vector associated with the second communication channel (block **840**). For example, the first network node (e.g., using communication manager **1108** and/or transmission component **1104**, depicted in FIG. **11**) may transmit the channel estimation information corresponding to the signal, wherein the channel estimation information includes first channel estimation information and second channel estimation information, wherein the first channel estimation information corresponds to the first communication channel and is based on the first pre-coding vector, and the second channel estimation information corresponds to the second communication channel and is based on a second pre-coding vector associated with the second communication channel, as described above.

[0120] Process **800** may include additional aspects, such as any single aspect or any combination of aspects described below and/or in connection with one or more other processes described elsewhere herein.

[0121] In some aspects, process **800** includes receiving at least one additional signal comprising at least one additional signal characteristic associated with the first communication channel and at least one additional signal characteristic associated with the second communication channel, and generating the channel estimation information based on a plurality of signals that includes the signal and the at least one additional signal. In some aspects, generating the channel estimation information comprises performing one or more channel estimation measurements based on the plurality of signals.

[0122] In some aspects, process **800** includes receiving CLI configuration information including first information indicative of a reference signal resource, wherein the signal corresponds to the reference signal resource. In some aspects, the CLI configuration information includes second information indicative of a plurality of pre-coding vectors corresponding to a pre-coding matrix, and the plurality of pre-coding vectors includes the first pre-coding vector and the second pre-coding vector. In some aspects, the plurality of pre-coding vectors includes at least one additional pre-coding vector associated with at least one additional network node configured to provide at least one additional electromagnetic radiation reflection relay service. In some aspects, the first network node comprises a memory that includes non-signaled information stored thereon, wherein the non-signaled information includes the first pre-coding vector and the second pre-coding vector.

[0123] In some aspects, the first channel estimation information is first raw channel estimation information, and the second channel estimation information is second raw channel estimation information, wherein the first raw channel estimation information and the second raw channel estimation information is non-differential information and non-quantized information. In some aspects, the first channel estimation information is indicative of a first channel estimation value associated with the first communication channel, and the second channel estimation information is a differential value indicative of a second channel estimation value, and the differential value is relative to the first channel estimation value. In some aspects, the first channel estimation information is first quantized channel estimation information, and the second channel estimation information is second quantized channel estimation information.

[0124] In some aspects, process **800** includes receiving CLI configuration information that specifies the second network node and the electromagnetic radiation reflection relay network node, and the second channel estimation information is based on the CLI configuration information.

[0125] Although FIG. **8** shows example blocks of process **800**, in some aspects, process **800** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. **8**. Additionally, or alternatively, two or more of the blocks of process **800** may be performed in parallel.

[0126] FIG. **9** is a diagram illustrating an example process **900** performed, for example, by a first network node, in accordance with the present disclosure. Example process **900** is an example where the first network node (e.g., the network node **702**) performs operations associated with reporting CLI associated with an electromagnetic radiation reflection relay service.

[0127] As shown in FIG. **9**, in some aspects, process **900** may include transmitting CLI configuration information that includes first information indicative of a reference signal resource (block **910**). For example, the first network node (e.g., using communication manager **1108** and/or transmission component **1104**, depicted in FIG. **11**) may transmit CLI configuration information that includes first information indicative of a reference signal resource, as described above.

[0128] As further shown in FIG. **9**, in some aspects, process **900** may include receiving channel estimation information corresponding to a signal that is based on the reference signal resource, wherein the channel estimation information includes first channel estimation information and second channel estimation information, wherein the first channel estimation information corresponds to a first communication channel between a second network node and a third network

node and is based on a first pre-coding vector associated with the first communication channel, and the second channel estimation information corresponds to a second communication channel between an electromagnetic radiation reflection relay network node and the third network node and is based on a second pre-coding vector associated with the second communication channel (block **920**). For example, the first network node (e.g., using communication manager **1108** and/or reception component **1102**, depicted in FIG. **11**) may receive channel estimation information corresponding to a signal that is based on the reference signal resource, wherein the channel estimation information includes first channel estimation information and second channel estimation information, wherein the first channel estimation information corresponds to a first communication channel between a second network node and a third network node and is based on a first pre-coding vector associated with the first communication channel, and the second channel estimation information corresponds to a second communication channel between an electromagnetic radiation reflection relay network node and the third network node and is based on a second pre-coding vector associated with the second communication channel, as described above.
[0129] As further shown in FIG. **9**, in some aspects, process **900** may include determining one or more electromagnetic reflection parameters based on the channel estimation information (block **930**). For example, the first network node (e.g., using communication manager **1108** and/or determination component **1112**, depicted in FIG. **11**) may determine one or more electromagnetic reflection parameters based on the channel estimation information, as described above.
[0130] As further shown in FIG. **9**, in some aspects, process **900** may include transmitting reflection relay configuration information to the electromagnetic radiation reflection relay network node, wherein the reflection relay configuration information is indicative of the one or more electromagnetic reflection parameters based on the channel estimation information (block **940**). For example, the first network node (e.g., using communication manager **1108** and/or transmission component **1104**, depicted in FIG. **11**) may transmit reflection relay configuration information to the electromagnetic radiation reflection relay network node, wherein the reflection relay configuration information is indicative of the one or more electromagnetic reflection parameters based on the channel estimation information, as described above.
[0131] Process **900** may include additional aspects, such as any single aspect or any combination of aspects described below and/or in connection with one or more other processes described elsewhere herein.
[0132] In some aspects, determining the one or more electromagnetic reflection parameters comprises determining at least one of a phase shift coefficient or an amplitude coefficient. In some aspects, determining the one or more electromagnetic reflection parameters comprises determining the one or more electromagnetic reflection parameters to mitigate CLI associated with at least one of the first communication channel or the second communication channel. In some aspects, determining the one or more electromagnetic reflection parameters comprises determining the one or more electromagnetic reflection parameters to increase a signal strength associated with the second communication channel. In some aspects, determining the one or more electromagnetic reflection parameters comprises determining the one or more electromagnetic reflection parameters to decrease a signal strength associated with the second communication channel.
[0133] In some aspects, the CLI configuration information includes second information indicative of a plurality of pre-coding vectors corresponding to a pre-coding matrix, and the plurality of pre-coding vectors includes the first pre-coding vector and the second pre-coding vector. In some aspects, the plurality of pre-coding vectors includes at least one additional pre-coding vector associated with at least one additional network node configured to provide at least one additional electromagnetic radiation reflection relay service. In some aspects, the first network node comprises a memory that includes non-signaled information stored thereon, wherein the non-signaled information includes the first pre-coding vector and the second pre-coding vector.
[0134] In some aspects, the first channel estimation information is first raw channel estimation

information, and the second channel estimation information is second raw channel estimation information, wherein the first raw channel estimation information and the second raw channel estimation information is non-differential information and non-quantized information. In some aspects, the first channel estimation information is indicative of a first channel estimation value associated with the first communication channel, and the second channel estimation information is a differential value indicative of a second channel estimation value, and the differential value is relative to the first channel estimation value. In some aspects, the first channel estimation information is first quantized channel estimation information, and the second channel estimation information is second quantized channel estimation information.

[0135] In some aspects, process **900** includes transmitting CLI configuration information that specifies the third network node and the electromagnetic radiation reflection relay network node, and the channel estimation information corresponds to the third network node and the electromagnetic radiation reflection relay network node based on the CLI configuration information.

[0136] Although FIG. **9** shows example blocks of process **900**, in some aspects, process **900** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. **9**. Additionally, or alternatively, two or more of the blocks of process **900** may be performed in parallel.

[0137] FIG. **10** is a diagram illustrating an example process **1000** performed, for example, by a first network node, in accordance with the present disclosure. Example process **1000** is an example where the first network node (e.g., the network node **704**) performs operations associated with reporting CLI associated with an electromagnetic radiation reflection relay service.

[0138] As shown in FIG. **10**, in some aspects, process **1000** may include receiving CLI configuration information (block **1010**). For example, the first network node (e.g., using communication manager **1108** and/or reception component **1102**, depicted in FIG. **11**) may receive CLI configuration information, as described above.

[0139] As further shown in FIG. **10**, in some aspects, process **1000** may include receiving, from an aggressor network node, a plurality of aggressor reference signals, wherein the plurality of aggressor reference signals includes a first aggressor reference signal pre-coded by first pre-coding information and a second aggressor reference signal pre-coded by second pre-coding information (block **1020**). For example, the first network node (e.g., using communication manager **1108** and/or reception component **1102**, depicted in FIG. **11**) may receive, from an aggressor network node, a plurality of aggressor reference signals, wherein the plurality of aggressor reference signals includes a first aggressor reference signal pre-coded by first pre-coding information and a second aggressor reference signal pre-coded by second pre-coding information, as described above.

[0140] As further shown in FIG. **10**, in some aspects, process **1000** may include receiving, from an electromagnetic radiation reflection relay network node, a plurality of reflected reference signals, wherein the plurality of reflected reference signals includes a first reflected reference signal pre-coded by third pre-coding information and a second reflected reference signal pre-coded by fourth pre-coding information, wherein the first aggressor reference signal and the first reflected reference signal overlap in time, and the second aggressor reference signal and the second reflected reference signal overlap in time (block **1030**). For example, the first network node (e.g., using communication manager **1108** and/or reception component **1102**, depicted in FIG. **11**) may receive, from an electromagnetic radiation reflection relay network node, a plurality of reflected reference signals, wherein the plurality of reflected reference signals includes a first reflected reference signal pre-coded by third pre-coding information and a second reflected reference signal pre-coded by fourth pre-coding information, wherein the first aggressor reference signal and the first reflected reference signal overlap in time, and the second aggressor reference signal and the second reflected reference signal overlap in time, as described above.

[0141] As further shown in FIG. **10**, in some aspects, process **1000** may include generating channel estimation information based on the plurality of aggressor reference signals and the plurality of reflected reference signals (block **1040**). For example, the first network node (e.g., using communication manager **1108** and/or generation component **1110**, depicted in FIG. **11**) may generate channel estimation information based on the plurality of aggressor reference signals and the plurality of reflected reference signals, as described above.

[0142] As further shown in FIG. **10**, in some aspects, process **1000** may include transmitting, to a second network node, the channel estimation information including first channel estimation information and second channel estimation information, wherein the channel estimation information is based on the plurality of aggressor reference signals and the plurality of reflected reference signals, wherein the first channel information is indicative of one or more channel characteristics corresponding to a first channel between the first network node and the aggressor network node, and the second channel information is indicative of one or more channel characteristics corresponding to a second channel between the first network node and the electromagnetic radiation reflection relay network node (block **1050**). For example, the first network node (e.g., using communication manager **1108** and/or transmission component **1104**, depicted in FIG. **11**) may transmit, to a second network node, the channel estimation information including first channel estimation information and second channel estimation information, wherein the channel estimation information is based on the plurality of aggressor reference signals and the plurality of reflected reference signals, wherein the first channel information is indicative of one or more channel characteristics corresponding to a first channel between the first network node and the aggressor network node, and the second channel information is indicative of one or more channel characteristics corresponding to a second channel between the first network node and the electromagnetic radiation reflection relay network node, as described above.

[0143] Process **1000** may include additional aspects, such as any single aspect or any combination of aspects described below and/or in connection with one or more other processes described elsewhere herein.

[0144] In some aspects, each respective reference signal of the plurality of aggressor reference signals comprises a respective sounding reference signal, and each respective reference signal of the plurality of reflected reference signals comprises a respective sounding reference signal. In some aspects, process **1000** includes generating the channel estimation information based on the plurality of aggressor reference signals and the plurality of reflected reference signals.

[0145] In some aspects, generating the channel estimation information comprises performing one or more channel estimation measurements based on the plurality of aggressor reference signals and the plurality of reflected reference signals. In some aspects, the CLI configuration information includes first information indicative of a plurality of reference signal resources, wherein the plurality of aggressor reference signals and the plurality of reflected reference signals correspond to the plurality of reference signal resources. In some aspects, the CLI configuration information includes second information indicative of a plurality of pre-coding vectors corresponding to a pre-coding matrix, and the plurality of pre-coding vectors includes the first pre-coding information, the second pre-coding information, the third pre-coding information, and the fourth pre-coding information. In some aspects, the plurality of pre-coding vectors includes at least one pre-coding vector corresponding to additional pre-coding information associated with at least one additional network node configured to provide at least one additional electromagnetic radiation reflection relay service.

[0146] In some aspects, the CLI configuration information specifies the aggressor network node and the electromagnetic radiation reflection relay network node, and the channel estimation information corresponds to the first channel and the second channel based on the CLI configuration information.

[0147] Although FIG. **10** shows example blocks of process **1000**, in some aspects, process **1000** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. **10**. Additionally, or alternatively, two or more of the blocks of process **1000** may be performed in parallel.

[0148] FIG. **11** is a diagram of an example apparatus **1100** for wireless communication, in accordance with the present disclosure. The apparatus **1100** may be a network node, or a network node may include the apparatus **1100**. In some aspects, the apparatus **1100** includes a reception component **1102** and a transmission component **1104**, which may be in communication with one another (for example, via one or more buses and/or one or more other components). As shown, the apparatus **1100** may communicate with another apparatus **1106** (such as a UE, a base station, or another wireless communication device) using the reception component **1102** and the transmission component **1104**. As further shown, the apparatus **1100** may include a communication manager **1108**. The communication manager **1108** may include one or more of a generation component **1110**, or a determination component **1112**, among other examples.

[0149] In some aspects, the apparatus **1100** may be configured to perform one or more operations described herein in connection with FIG. **7**. Additionally, or alternatively, the apparatus **1100** may be configured to perform one or more processes described herein, such as process **800** of FIG. **8**, process **900** of FIG. **9**, process **1000** of FIG. **10**, or a combination thereof. In some aspects, the apparatus **1100** and/or one or more components shown in FIG. **11** may include one or more components of one or more of the network nodes described in connection with FIG. **4**. Additionally, or alternatively, one or more components shown in FIG. **11** may be implemented within one or more components described in connection with FIG. **4**. Additionally, or alternatively, one or more components of the set of components may be implemented at least in part as software stored in a memory. For example, a component (or a portion of a component) may be implemented as instructions or code stored in a non-transitory computer-readable medium and executable by a controller or a processor to perform the functions or operations of the component.

[0150] The reception component **1102** may receive communications, such as reference signals, control information, data communications, or a combination thereof, from the apparatus **1106**. The reception component **1102** may provide received communications to one or more other components of the apparatus **1100**. In some aspects, the reception component **1102** may perform signal processing on the received communications (such as filtering, amplification, demodulation, analog-to-digital conversion, demultiplexing, deinterleaving, de-mapping, equalization, interference cancellation, or decoding, among other examples), and may provide the processed signals to the one or more other components of the apparatus **1100**. In some aspects, the reception component **1102** may include one or more antennas, a modem, a demodulator, a MIMO detector, a receive processor, a controller/processor, a memory, or a combination thereof, of one or more of the network nodes described in connection with FIG. **4**.

[0151] The transmission component **1104** may transmit communications, such as reference signals, control information, data communications, or a combination thereof, to the apparatus **1106**. In some aspects, one or more other components of the apparatus **1100** may generate communications and may provide the generated communications to the transmission component **1104** for transmission to the apparatus **1106**. In some aspects, the transmission component **1104** may perform signal processing on the generated communications (such as filtering, amplification, modulation, digital-to-analog conversion, multiplexing, interleaving, mapping, or encoding, among other examples), and may transmit the processed signals to the apparatus **1106**. In some aspects, the transmission component **1104** may include one or more antennas, a modem, a modulator, a transmit MIMO processor, a transmit processor, a controller/processor, a memory, or a combination thereof, of one or more of the network nodes described in connection with FIG. **4**. In some aspects, the transmission component **1104** may be co-located with the reception component **1102** in a transceiver.

[0152] The communication manager **1108** and/or the reception component **1102** may receive CLI configuration information. In some aspects, the communication manager **1108** may include one or more antennas, a modem, a controller/processor, a memory, or a combination thereof, of one or more of the network nodes described in connection with FIG. **4**. In some aspects, the communication manager **1108** may include the reception component **1102** and/or the transmission component **1104**. In some aspects, the communication manager **1108** may be, be similar to, include, or be included in, the communication manager **340** and/or **350** depicted in FIGS. **3** and **4**.

[0153] The communication manager **1108** and/or the reception component **1102** may receive a signal comprising a first signal characteristic associated with a first pre-coding vector corresponding to a first communication channel between the first network node and a second network node and a second signal characteristic associated with a second pre-coding vector corresponding to a second communication channel between the first network node and an electromagnetic radiation reflection relay network node.

[0154] The communication manager **1108** and/or the generation component **1110** may generate channel estimation information corresponding to the signal. In some aspects, the generation component **1110** may include one or more antennas, a modem, a controller/processor, a memory, or a combination thereof, of one or more of the network nodes described in connection with FIG. **4**. In some aspects, the generation component **1110** may include the reception component **1102** and/or the transmission component **1104**.

[0155] The communication manager **1108** and/or the transmission component **1104** may transmit the channel estimation information corresponding to the signal, wherein the channel estimation information includes first channel estimation information and second channel estimation information, wherein the first channel estimation information corresponds to the first communication channel and is based on the first pre-coding vector, and the second channel estimation information corresponds to the second communication channel and is based on a second pre-coding vector associated with the second communication channel. The communication manager **1108** and/or the reception component **1102** may receive at least one additional signal comprising at least one additional signal characteristic associated with the first communication channel and at least one additional signal characteristic associated with the second communication channel.

[0156] The communication manager **1108** and/or the generation component **1110** may generate the channel estimation information based on a plurality of signals that includes the signal and the at least one additional signal. The communication manager **1108** and/or the reception component **1102** may receive CLI configuration information including first information indicative of a reference signal resource, wherein the signal corresponds to the reference signal resource. The communication manager **1108** and/or the reception component **1102** may receive CLI configuration information that specifies the second network node and the electromagnetic radiation reflection relay network node, and the second channel estimation information is based on the CLI configuration information.

[0157] The communication manager **1108** and/or the transmission component **1104** may transmit CLI configuration information that includes first information indicative of a reference signal resource. The reception component **1102** may receive channel estimation information corresponding to a signal that is based on the reference signal resource, wherein the channel estimation information includes first channel estimation information and second channel estimation information, wherein the first channel estimation information corresponds to a first communication channel between a second network node and a third network node and is based on a first pre-coding vector associated with the first communication channel, and the second channel estimation information corresponds to a second communication channel between an electromagnetic radiation reflection relay network node and the third network node and is based on a second pre-coding vector associated with the second communication channel.

[0158] The communication manager **1108** and/or the determination component **1112** may determine one or more electromagnetic reflection parameters based on the channel estimation information. In some aspects, the determination component **1112** may include one or more antennas, a modem, a controller/processor, a memory, or a combination thereof, of one or more of the network nodes described in connection with FIG. **4**. In some aspects, the determination component **1112** may include the reception component **1102** and/or the transmission component **1104**.

[0159] The communication manager **1108** and/or the transmission component **1104** may transmit reflection relay configuration information to the electromagnetic radiation reflection relay network node, wherein the reflection relay configuration information is indicative of the one or more electromagnetic reflection parameters based on the channel estimation information. The communication manager **1108** and/or the transmission component **1104** may transmit CLI configuration information that specifies the third network node and the electromagnetic radiation reflection relay network node, and the channel estimation information corresponds to the third network node and the electromagnetic radiation reflection relay network node based on the CLI configuration information.

[0160] The communication manager **1108** and/or the reception component **1102** may receive CLI configuration information. The communication manager **1108** and/or the reception component **1102** may receive, from an aggressor network node, a plurality of aggressor reference signals, wherein the plurality of aggressor reference signals includes a first aggressor reference signal pre-coded by first pre-coding information and a second aggressor reference signal pre-coded by second pre-coding information. The communication manager **1108** and/or the reception component **1102** may receive, from an electromagnetic radiation reflection relay network node, a plurality of reflected reference signals, wherein the plurality of reflected reference signals includes a first reflected reference signal pre-coded by third pre-coding information and a second reflected reference signal pre-coded by fourth pre-coding information, wherein the first aggressor reference signal and the first reflected reference signal overlap in time, and the second aggressor reference signal and the second reflected reference signal overlap in time.

[0161] The communication manager **1108** and/or the generation component **1110** may generate channel estimation information based on the plurality of aggressor reference signals and the plurality of reflected reference signals. The communication manager **1108** and/or the transmission component **1104** may transmit, to a second network node, the channel estimation information including first channel estimation information and second channel estimation information, wherein the channel estimation information is based on the plurality of aggressor reference signals and the plurality of reflected reference signals, wherein the first channel information is indicative of one or more channel characteristics corresponding to a first channel between the first network node and the aggressor network node, and the second channel information is indicative of one or more channel characteristics corresponding to a second channel between the first network node and the electromagnetic radiation reflection relay network node. The communication manager **1108** and/or the generation component **1110** may generate the channel estimation information based on the plurality of aggressor reference signals and the plurality of reflected reference signals.

[0162] The number and arrangement of components shown in FIG. **11** are provided as an example. In practice, there may be additional components, fewer components, different components, or differently arranged components than those shown in FIG. **11**. Furthermore, two or more components shown in FIG. **11** may be implemented within a single component, or a single component shown in FIG. **11** may be implemented as multiple, distributed components. Additionally, or alternatively, a set of (one or more) components shown in FIG. **11** may perform one or more functions described as being performed by another set of components shown in FIG. **11**.

[0163] The following provides an overview of some Aspects of the present disclosure:

[0164] Aspect 1: A method of wireless communication performed by a first network node, comprising: receiving a signal comprising a first signal characteristic associated with a first pre-coding vector corresponding to a first communication channel between the first network node and a second network node and a second signal characteristic associated with a second pre-coding vector corresponding to a second communication channel between the first network node and an electromagnetic radiation reflection relay network node; and transmitting channel estimation information corresponding to the signal, wherein the channel estimation information includes first channel estimation information and second channel estimation information, wherein the first channel estimation information corresponds to the first communication channel and is based on the first pre-coding vector, and wherein the second channel estimation information corresponds to the second communication channel and is based on a second pre-coding vector associated with the second communication channel.

[0165] Aspect 2: The method of Aspect 1, wherein the signal comprises a reference signal.

[0166] Aspect 3: The method of Aspect 2, wherein the reference signal comprises a sounding reference signal.

[0167] Aspect 4: The method of any of Aspects 1-3, further comprising: receiving at least one additional signal comprising at least one additional signal characteristic associated with the first communication channel and at least one additional signal characteristic associated with the second communication channel; and generating the channel estimation information based on a plurality of signals that includes the signal and the at least one additional signal.

[0168] Aspect 5: The method of Aspect 4, wherein generating the channel estimation information comprises performing one or more channel estimation measurements based on the plurality of signals.

[0169] Aspect 6: The method of any of Aspects 1-5, further comprising receiving cross-link interference (CLI) configuration information including first information indicative of a reference signal resource, wherein the signal corresponds to the reference signal resource.

[0170] Aspect 7: The method of Aspect 6, wherein the CLI configuration information includes second information indicative of a plurality of pre-coding vectors corresponding to a pre-coding matrix, and wherein the plurality of pre-coding vectors includes the first pre-coding vector and the second pre-coding vector.

[0171] Aspect 8: The method of Aspect 7, wherein the plurality of pre-coding vectors includes at least one additional pre-coding vector associated with at least one additional network node configured to provide at least one additional electromagnetic radiation reflection relay service.

[0172] Aspect 9: The method of any of Aspects 1-8, wherein the first network node comprises a memory that includes non-signaled information stored thereon, wherein the non-signaled information includes the first pre-coding vector and the second pre-coding vector.

[0173] Aspect 10: The method of any of Aspects 1-9, wherein the first channel estimation information is first raw channel estimation information and the second channel estimation information is second raw channel estimation information, wherein the first raw channel estimation information and the second raw channel estimation information is non-differential information and non-quantized information.

[0174] Aspect 11: The method of any of Aspects 1-9, wherein the first channel estimation information is indicative of a first channel estimation value associated with the first communication channel, and the second channel estimation information is a differential value indicative of a second channel estimation value, and wherein the differential value is relative to the first channel estimation value.

[0175] Aspect 12: The method of any of Aspects 1-9 or 11, wherein the first channel estimation information is first quantized channel estimation information, and wherein the second channel estimation information is second quantized channel estimation information.

[0176] Aspect 13: The method of any of Aspects 1-12, further comprising receiving cross-link interference (CLI) configuration information that specifies the second network node and the electromagnetic radiation reflection relay network node, and wherein the second channel estimation information is based on the CLI configuration information.

[0177] Aspect 14: A method of wireless communication performed by a first network node, comprising: transmitting cross-link interference (CLI) configuration information that includes first information indicative of a reference signal resource; receiving channel estimation information corresponding to a signal that is based on the reference signal resource, wherein the channel estimation information includes first channel estimation information and second channel estimation information, wherein the first channel estimation information corresponds to a first communication channel between a second network node and a third network node and is based on a first pre-coding vector associated with the first communication channel, and wherein the second channel estimation information corresponds to a second communication channel between an electromagnetic radiation reflection relay network node and the third network node and is based on a second pre-coding vector associated with the second communication channel; and transmitting reflection relay configuration information to the electromagnetic radiation reflection relay network node, wherein the reflection relay configuration information is indicative of one or more electromagnetic reflection parameters based on the channel estimation information.

[0178] Aspect 15: The method of Aspect 14, further comprising determining the one or more electromagnetic reflection parameters based on the channel estimation information.

[0179] Aspect 16: The method of either of Aspects 14 or 15, wherein determining the one or more electromagnetic reflection parameters comprises determining at least one of a phase shift coefficient or an amplitude coefficient.

[0180] Aspect 17: The method of any of Aspects 14-16, wherein determining the one or more electromagnetic reflection parameters comprises determining the one or more electromagnetic reflection parameters to mitigate CLI associated with at least one of the first communication channel or the second communication channel.

[0181] Aspect 18: The method of any of Aspects 14-17, wherein determining the one or more electromagnetic reflection parameters comprises determining the one or more electromagnetic reflection parameters to increase a signal strength associated with the second communication channel.

[0182] Aspect 19: The method of any of Aspects 14-17, wherein determining the one or more electromagnetic reflection parameters comprises determining the one or more electromagnetic reflection parameters to decrease a signal strength associated with the second communication channel.

[0183] Aspect 20: The method of any of Aspects 14-19, wherein the CLI configuration information includes second information indicative of a plurality of pre-coding vectors corresponding to a pre-coding matrix, and wherein the plurality of pre-coding vectors includes the first pre-coding vector and the second pre-coding vector.

[0184] Aspect 21: The method of Aspect 20, wherein the plurality of pre-coding vectors includes at least one additional pre-coding vector associated with at least one additional network node configured to provide at least one additional electromagnetic radiation reflection relay service.

[0185] Aspect 22: The method of any of Aspects 14-21, wherein the first network node comprises a memory that includes non-signaled information stored thereon, wherein the non-signaled information includes the first pre-coding vector and the second pre-coding vector.

[0186] Aspect 23: The method of any of Aspects 14-22, wherein the first channel estimation information is first raw channel estimation information and the second channel estimation information is second raw channel estimation information, wherein the first raw channel estimation information and the second raw channel estimation information is non-differential information and non-quantized information.

[0187] Aspect 24: The method of any of Aspects 14-22, wherein the first channel estimation information is indicative of a first channel estimation value associated with the first communication channel, and the second channel estimation information is a differential value indicative of a second channel estimation value, and wherein the differential value is relative to the first channel estimation value.

[0188] Aspect 25: The method of any of Aspects 14-22 or 23, wherein the first channel estimation information is first quantized channel estimation information, and wherein the second channel estimation information is second quantized channel estimation information.

[0189] Aspect 26: The method of any of Aspects 14-25, further comprising transmitting cross-link interference (CLI) configuration information that specifies the third network node and the electromagnetic radiation reflection relay network node, and wherein the channel estimation information corresponds to the third network node and the electromagnetic radiation reflection relay network node based on the CLI configuration information.

[0190] Aspect 27: A method of wireless communication performed by a first network node, comprising: receiving, from an aggressor network node, a plurality of aggressor reference signals, wherein the plurality of aggressor reference signals includes a first aggressor reference signal pre-coded by first pre-coding information and a second aggressor reference signal pre-coded by second pre-coding information; receiving, from an electromagnetic radiation reflection relay network node, a plurality of reflected reference signals, wherein the plurality of reflected reference signals includes a first reflected reference signal pre-coded by third pre-coding information and a second reflected reference signal pre-coded by fourth pre-coding information, wherein the first aggressor reference signal and the first reflected reference signal overlap in time, and the second aggressor reference signal and the second reflected reference signal overlap in time; and transmitting, to a second network node, channel estimation information including first channel estimation information and second channel estimation information, wherein the channel estimation information is based on the plurality of aggressor reference signals and the plurality of reflected reference signals, wherein the first channel information is indicative of one or more channel characteristics corresponding to a first channel between the first network node and the aggressor network node, and wherein the second channel information is indicative of one or more channel characteristics corresponding to a second channel between the first network node and the electromagnetic radiation reflection relay network node.

[0191] Aspect 28: The method of Aspect 27, wherein each respective reference signal of the plurality of aggressor reference signals comprises a respective sounding reference signal, and wherein each respective reference signal of the plurality of reflected reference signals comprises a respective sounding reference signal.

[0192] Aspect 29: The method of either of Aspects 27 or 28, further comprising generating the channel estimation information based on the plurality of aggressor reference signals and the plurality of reflected reference signals.

[0193] Aspect 30: The method of Aspect 29, wherein generating the channel estimation information comprises performing one or more channel estimation measurements based on the plurality of aggressor reference signals and the plurality of reflected reference signals.

[0194] Aspect 31: The method of any of Aspects 27-30, wherein further comprising receiving cross-link interference (CLI) configuration information including first information indicative of a plurality of reference signal resources, wherein the plurality of aggressor reference signals and the plurality of reflected reference signals correspond to the plurality of reference signal resources.

[0195] Aspect 32: The method of Aspect 31, wherein the CLI configuration information includes second information indicative of a plurality of pre-coding vectors corresponding to a pre-coding matrix, and wherein the plurality of pre-coding vectors includes the first pre-coding information, the second pre-coding information, the third pre-coding information, and the fourth pre-coding information.

[0196] Aspect 33: The method of Aspect 32, wherein the plurality of pre-coding vectors includes at least one pre-coding vector corresponding to additional pre-coding information associated with at least one additional network node configured to provide at least one additional electromagnetic

radiation reflection relay service.
[0197] Aspect 34: The method of any of Aspects 27-33, further comprising receiving cross-link interference (CLI) configuration information that specifies the aggressor network node and the electromagnetic radiation reflection relay network node, and wherein the channel estimation information corresponds to the first channel and the second channel based on the CLI configuration information.
[0198] Aspect 35: An apparatus for wireless communication at a device, comprising a processor; memory coupled with the processor; and instructions stored in the memory and executable by the processor to cause the apparatus to perform the method of one or more of Aspects 1-13.
[0199] Aspect 36: A device for wireless communication, comprising a memory and one or more processors coupled to the memory, the one or more processors configured to perform the method of one or more of Aspects 1-13.
[0200] Aspect 37: An apparatus for wireless communication, comprising at least one means for performing the method of one or more of Aspects 1-13.
[0201] Aspect 38: A non-transitory computer-readable medium storing code for wireless communication, the code comprising instructions executable by a processor to perform the method of one or more of Aspects 1-13.
[0202] Aspect 39: A non-transitory computer-readable medium storing a set of instructions for wireless communication, the set of instructions comprising one or more instructions that, when executed by one or more processors of a device, cause the device to perform the method of one or more of Aspects 1-13.
[0203] Aspect 40: An apparatus for wireless communication at a device, comprising a processor; memory coupled with the processor; and instructions stored in the memory and executable by the processor to cause the apparatus to perform the method of one or more of Aspects 14-26.
[0204] Aspect 41: A device for wireless communication, comprising a memory and one or more processors coupled to the memory, the one or more processors configured to perform the method of one or more of Aspects 14-26.
[0205] Aspect 42: An apparatus for wireless communication, comprising at least one means for performing the method of one or more of Aspects 14-26.
[0206] Aspect 43: A non-transitory computer-readable medium storing code for wireless communication, the code comprising instructions executable by a processor to perform the method of one or more of Aspects 14-26.
[0207] Aspect 44: A non-transitory computer-readable medium storing a set of instructions for wireless communication, the set of instructions comprising one or more instructions that, when executed by one or more processors of a device, cause the device to perform the method of one or more of Aspects 14-26.
[0208] Aspect 45: An apparatus for wireless communication at a device, comprising a processor; memory coupled with the processor; and instructions stored in the memory and executable by the processor to cause the apparatus to perform the method of one or more of Aspects 27-34.
[0209] Aspect 46: A device for wireless communication, comprising a memory and one or more processors coupled to the memory, the one or more processors configured to perform the method of one or more of Aspects 27-34.
[0210] Aspect 47: An apparatus for wireless communication, comprising at least one means for performing the method of one or more of Aspects 27-34.
[0211] Aspect 48: A non-transitory computer-readable medium storing code for wireless communication, the code comprising instructions executable by a processor to perform the method of one or more of Aspects 27-34.
[0212] Aspect 49: A non-transitory computer-readable medium storing a set of instructions for wireless communication, the set of instructions comprising one or more instructions that, when executed by one or more processors of a device, cause the device to perform the method of one or more or

more of Aspects 27-34.
[0213] The foregoing disclosure provides illustration and description but is not intended to be exhaustive or to limit the aspects to the precise forms disclosed. Modifications and variations may be made in light of the above disclosure or may be acquired from practice of the aspects.
[0214] As used herein, the term “component” is intended to be broadly construed as hardware and/or a combination of hardware and software. “Software” shall be construed broadly to mean instructions, instruction sets, code, code segments, program code, programs, subprograms, software modules, applications, software applications, software packages, routines, subroutines, objects, executables, threads of execution, procedures, and/or functions, among other examples, whether referred to as software, firmware, middleware, microcode, hardware description language, or otherwise. As used herein, a “processor” is implemented in hardware and/or a combination of hardware and software. It will be apparent that systems and/or methods described herein may be implemented in different forms of hardware and/or a combination of hardware and software. The actual specialized control hardware or software code used to implement these systems and/or methods is not limiting of the aspects. Thus, the operation and behavior of the systems and/or methods are described herein without reference to specific software code, since those skilled in the art will understand that software and hardware can be designed to implement the systems and/or methods based, at least in part, on the description herein.
[0215] As used herein, “satisfying a threshold” may, depending on the context, refer to a value being greater than the threshold, greater than or equal to the threshold, less than the threshold, less than or equal to the threshold, equal to the threshold, not equal to the threshold, or the like.
[0216] Even though particular combinations of features are recited in the claims and/or disclosed in the specification, these combinations are not intended to limit the disclosure of various aspects. Many of these features may be combined in ways not specifically recited in the claims and/or disclosed in the specification. The disclosure of various aspects includes each dependent claim in combination with every other claim in the claim set. As used herein, a phrase referring to “at least one of” a list of items refers to any combination of those items, including single members. As an example, “at least one of: a, b, or c” is intended to cover a, b, c, a+b, a+c, b+c, and a+b+c, as well as any combination with multiples of the same element (e.g., a+a, a+a+a, a+a+b, a+a+c, a+b+b, a+c+c, b+b, b+b+b, b+b+c, c+c, and c+c+c, or any other ordering of a, b, and c).
[0217] No element, act, or instruction used herein should be construed as critical or essential unless explicitly described as such. Also, as used herein, the articles “a” and “an” are intended to include one or more items and may be used interchangeably with “one or more.” Further, as used herein, the article “the” is intended to include one or more items referenced in connection with the article “the” and may be used interchangeably with “the one or more.” Furthermore, as used herein, the terms “set” and “group” are intended to include one or more items and may be used interchangeably with “one or more.” Where only one item is intended, the phrase “only one” or similar language is used. Also, as used herein, the terms “has,” “have,” “having,” or the like are intended to be open-ended terms that do not limit an element that they modify (e.g., an element “having” A may also have B). As used herein, the phrase “based on” shall not be construed as a reference to a closed set of information, one or more conditions, one or more factors, or the like. In other words, the phrase “based on A” (where “A” may be information, a condition, a factor, or the like) shall be construed as “based at least on A” unless specifically recited differently.

Claims

1. A first network node for wireless communication, comprising: a memory; and at least one processor coupled to the memory, wherein the at least one processor is configured to: receive a signal comprising a first signal characteristic associated with a first pre-coding vector corresponding to a first communication channel between the first network node and a second network node and a second signal characteristic associated with a second pre-coding vector corresponding to a second communication channel between the first network node and an electromagnetic radiation reflection relay network node; and transmit channel estimation information corresponding to the signal, wherein the channel estimation information includes first channel estimation information and second channel estimation information, wherein the first channel estimation information corresponds to the first communication channel and is based on the first pre-coding vector, and wherein the second channel estimation information corresponds to the second communication channel and is based on a second pre-coding vector associated with the second communication channel.

2. The first network node of claim 1, wherein the signal comprises a reference signal.

3. The first network node of claim 2, wherein the reference signal comprises a sounding reference signal.

4. The first network node of claim 1, wherein the at least one processor is further configured to: receive at least one additional signal comprising at least one additional signal characteristic associated with the first communication channel and at least one additional signal characteristic associated with the second communication channel; and generate the channel estimation information based on a plurality of signals that includes the signal and the at least one additional signal.

5. The first network node of claim 4, wherein the at least one processor, to generate the channel estimation information, is configured to perform one or more channel estimation measurements based on the plurality of signals.

6. The first network node of claim 1, wherein the at least one processor is further configured to receive cross-link interference (CLI) configuration information including first information indicative of a reference signal resource, wherein the signal corresponds to the reference signal resource.

7. The first network node of claim 6, wherein the CLI configuration information includes second information indicative of a plurality of pre-coding vectors corresponding to a pre-coding matrix, and wherein the plurality of pre-coding vectors includes the first pre-coding vector and the second pre-coding vector.

8. The first network node of claim 7, wherein the plurality of pre-coding vectors includes at least one additional pre-coding vector associated with at least one additional network node configured to provide at least one additional electromagnetic radiation reflection relay service.

9. The first network node of claim 1, wherein the memory includes non-signaled information stored thereon, wherein the non-signaled information includes the first pre-coding vector and the second pre-coding vector.

10. The first network node of claim 1, wherein the first channel estimation information is first raw channel estimation information and the second channel estimation information is second raw channel estimation information, wherein the first raw channel estimation information and the second raw channel estimation information is non-differential information and non-quantized information.

11. The first network node of claim 1, wherein the first channel estimation information is indicative of a first channel estimation value associated with the first communication channel, and the second channel estimation information is a differential value indicative of a second channel estimation value, and wherein the differential value is relative to the first channel estimation value.

12. The first network node of claim 1, wherein the first channel estimation information is first quantized channel estimation information, and wherein the second channel estimation information is second quantized channel estimation information.

13. The first network node of claim 1, wherein the at least one processor is further configured to receive cross-link interference (CLI) configuration information that specifies the second network node and the electromagnetic radiation reflection relay network node, and wherein the second channel estimation information is based on the CLI configuration information.

14. A first network node for wireless communication, comprising: a memory; and at least one processor coupled to the memory, wherein the at least one processor is configured to: transmit cross-link interference (CLI) configuration information that includes first information indicative of a reference signal resource; receive channel estimation information corresponding to a signal that is based on the reference signal resource, wherein the channel estimation information includes first channel estimation information and second channel estimation information, wherein the first channel estimation information corresponds to a first communication channel between a second network node and a third network node and is based on a first pre-coding vector associated with the first communication channel, and wherein the second channel estimation information corresponds to a second communication channel between an electromagnetic radiation reflection relay network node and the third network node and is based on a second pre-coding vector associated with the second communication channel; and transmit reflection relay configuration information to the electromagnetic radiation reflection relay network node, wherein the reflection relay configuration information is indicative of one or more electromagnetic reflection parameters based on the channel estimation information.

15. The first network node of claim 14, wherein the at least one processor is further configured to determine the one or more electromagnetic reflection parameters based on the channel estimation information.

16. The first network node of claim 14, wherein the at least one processor, to determine the one or more electromagnetic reflection parameters, is configured to determine at least one of a phase shift coefficient or an amplitude coefficient.

17. The first network node of claim 14, wherein the at least one processor, to determine the one or more electromagnetic reflection parameters, is configured to determine the one or more electromagnetic reflection parameters to mitigate CLI associated with at least one of the first communication channel or the second communication channel.

18. The first network node of claim 14, wherein the at least one processor, to determine the one or more electromagnetic reflection parameters, is configured to determine the one or more electromagnetic reflection parameters to increase a signal strength associated with the second communication channel.

19. The first network node of claim 14, wherein the at least one processor, to determine the one or more electromagnetic reflection parameters, is configured to determine the one or more electromagnetic reflection parameters to decrease a signal strength associated with the second communication channel.

20. The first network node of claim 14, wherein the CLI configuration information includes second information indicative of a plurality of pre-coding vectors corresponding to a pre-coding matrix, and wherein the plurality of pre-coding vectors includes the first pre-coding vector and the second pre-coding vector.

21. The first network node of claim 20, wherein the plurality of pre-coding vectors includes at least one additional pre-coding vector associated with at least one additional network node configured to provide at least one additional electromagnetic radiation reflection relay service.

22. The first network node of claim 14, wherein the memory includes non-signaled information stored thereon, wherein the non-signaled information includes the first pre-coding vector and the second pre-coding vector.

23. The first network node of claim 14, wherein the first channel estimation information is first raw channel estimation information and the second channel estimation information is second raw channel estimation information, wherein the first raw channel estimation information and the second raw channel estimation information is non-differential information and non-quantized information.

24. The first network node of claim 14, wherein the first channel estimation information is indicative of a first channel estimation value associated with the first communication channel, and the second channel estimation information is a differential value indicative of a second channel estimation value, and wherein the differential value is relative to the first channel estimation value.

25. The first network node of claim 14, wherein the first channel estimation information is first quantized channel estimation information, and wherein the second channel estimation information is second quantized channel estimation information.

26. The first network node of claim 14, wherein the at least one processor is further configured to transmit cross-link interference (CLI) configuration information that specifies the third network node and the electromagnetic radiation reflection relay network node, and wherein the channel estimation information corresponds to the third network node and the electromagnetic radiation reflection relay network node based on the CLI configuration information.

27. A first network node for wireless communication, comprising: a memory; and at least one processor coupled to the memory, wherein the at least one processor is configured to: receive, from an aggressor network node, a plurality of aggressor reference signals, wherein the plurality of aggressor reference signals includes a first aggressor reference signal pre-coded by first pre-coding information and a second aggressor reference signal pre-coded by second pre-coding information; receive, from an electromagnetic radiation reflection relay network node, a plurality of reflected reference signals, wherein the plurality of reflected reference signals includes a first reflected reference signal pre-coded by third pre-coding information and a second reflected reference signal pre-coded by fourth pre-coding information, wherein the first aggressor reference signal and the first reflected reference signal overlap in time, and the second aggressor reference signal and the second reflected reference signal overlap in time; and transmit, to a second network node, channel estimation information including first channel estimation information and second channel estimation information, wherein the channel estimation information is based on the plurality of aggressor reference signals and the plurality of reflected reference signals, wherein the first channel information is indicative of one or more channel characteristics corresponding to a first channel between the first network node and the aggressor network node, and wherein the second channel information is indicative of one or more channel characteristics corresponding to a second channel between the first network node and the electromagnetic radiation reflection relay network node.

28. The first network node of claim 27, wherein each respective reference signal of the plurality of aggressor reference signals comprises a respective sounding reference signal, and wherein each respective reference signal of the plurality of reflected reference signals comprises a respective sounding reference signal.

29. The first network node of claim 27, wherein the at least one processor is configured to generate the channel estimation information based on the plurality of aggressor reference signals and the plurality of reflected reference signals.

30. The first network node of claim 27, wherein the at least one processor is configured to receive cross-link interference (CLI) configuration information including first information indicative of a plurality of reference signal resources, wherein the plurality of aggressor reference signals and the plurality of reflected reference signals correspond to the plurality of reference signal resources, wherein the CLI configuration information includes second information indicative of a plurality of pre-coding vectors corresponding to a pre-coding matrix, and wherein the plurality of pre-coding vectors includes the first pre-coding information, the second pre-coding information, the third pre-coding information, and the fourth pre-coding information.